

Singular Moduli and the Ideal Class Group

A Lean 4 formalization blueprint

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July 20, 2026

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Chapter 1

Notation and standing conventions

Definition 1.1 (Notation and standing conventions).

Lean: QuadraticOrder.IsDiscriminant, QuadraticOrder.
IsFundamentalDiscriminant, QuadraticOrder.ConductorPrimeSetup,
QuadraticOrder.valD4, QuadraticOrder.epsilonEven, QuadraticOrder.
kroneckerAtPrime, QuadraticOrder.normForm, QuadraticOrder.tauConj,
QuadraticOrder.normEval, QuadraticOrder.idealIndex, QuadraticOrder.
idealCount, QuadraticOrder.IsInvertibleIdeal, QuadraticOrder.
invertibleIdealCount, QuadraticOrder.rootCountMod

Throughout the Chapter 3 development the following notation is in force.

- (N1) **Discriminants.** An integer d is a *discriminant* if $d \equiv 0$ or $1 \pmod{4}$; an *imaginary quadratic discriminant* if moreover $d < 0$. The letter d always denotes an imaginary quadratic discriminant.
- (N2) **Fundamental discriminant and conductor.** A discriminant D is *fundamental* if either $D \equiv 1 \pmod{4}$ and D is squarefree, or $D = 4m$ with m squarefree and $m \equiv 2, 3 \pmod{4}$. Every imaginary quadratic discriminant factors uniquely as $d = Df^2$ with D fundamental and $f \in \mathbb{Z}_{\geq 1}$, the *conductor*. We write $K := \mathbb{Q}(\sqrt{d}) = \mathbb{Q}(\sqrt{D})$, $\mathcal{O}_K = \mathbb{Z}[\omega]$ with $\omega := \frac{D+\sqrt{D}}{2}$, so $D = \text{disc}(K)$.
- (N3) **The order.** Set $g(x) := x^2 - dx + \frac{d^2-d}{4} \in \mathbb{Z}[x]$ and $\tau := \frac{d+\sqrt{d}}{2}$ (square root with positive imaginary part); g is the minimal polynomial of τ and $\text{disc}(g) = d$. Define $\mathcal{O}_d := \mathbb{Z}[\tau] \cong \mathbb{Z}[x]/(g(x))$, free over $\mathbb{Z}$ with basis $(1, \tau)$. With $d = Df^2$ one has $\tau = f\omega + \frac{Df(f-1)}{2}$, hence $\mathcal{O}_d = \mathbb{Z} + f\mathcal{O}_K$ and $[\mathcal{O}_K : \mathcal{O}_d] = f$. We write $\mathcal{O}$ for $\mathcal{O}_d$ when d is clear.

- (N4) **Conjugation and norm form.** $\bar{\tau} := d - \tau$ (the other root of g); $\alpha \mapsto \bar{\alpha}$ is the ring involution of $\mathcal{O}_d$ fixing $\mathbb{Z}$ with $\tau \mapsto \bar{\tau}$. Vieta: $\tau + \bar{\tau} = d$, $\tau\bar{\tau} = \frac{d^2-d}{4}$, $(\tau - \bar{\tau})^2 = d$. The norm is $N(\alpha) := \alpha\bar{\alpha}$; in coordinates

$$N(x + y\tau) = x^2 + dxy + \frac{d^2-d}{4}y^2,$$

the homogenization of g (for $y \neq 0$, $N(x + y\tau) = y^2g(-x/y)$; in particular $N(x - \tau) = g(x)$). It is multiplicative and, since $d < 0$, positive definite: $N(\alpha) > 0$ for $\alpha \neq 0$.

- (N5) **Ideal index — the norm convention.** For an ideal $\mathfrak{a} \subseteq \mathcal{O}_d$, $[\mathcal{O}_d : \mathfrak{a}] := \#(\mathcal{O}_d/\mathfrak{a})$, with value 0 when the quotient is infinite (only for $\mathfrak{a} = (0)$); every nonzero ideal has finite index. *The norm of an ideal is by definition its index*, $N(\mathfrak{a}) := [\mathcal{O}_d : \mathfrak{a}]$, for all nonzero ideals, invertible or not; “ideal of norm n ” always means $[\mathcal{O}_d : \mathfrak{a}] = n$. On principal ideals the conventions agree: $[\mathcal{O}_d : \alpha\mathcal{O}_d] = N(\alpha)$ for $\alpha \neq 0$.

- (N6) **Invertible ideals.** An ideal $\mathfrak{a} \subseteq \mathcal{O}_d$ is *invertible* if there exist an ideal $\mathfrak{b} \subseteq \mathcal{O}_d$ and $x \in \mathcal{O}_d \setminus \{0\}$ with $\mathfrak{a}\mathfrak{b} = x\mathcal{O}_d$. (Equivalence with local principality is proved in the invertibility-interface nodes, not assumed here.) $\text{Cl}(\mathcal{O}_d) = \text{Pic}(\mathcal{O}_d)$ denotes invertible ideals modulo nonzero principal ideals, $h_d := \# \text{Cl}(\mathcal{O}_d)$.

- (N7) **Counting functions.** For $n \geq 1$,

$$r_d(n) := \#\{\mathfrak{a} \subseteq \mathcal{O}_d \text{ ideal} : [\mathcal{O}_d : \mathfrak{a}] = n\}, \quad R_d(n) := \#\{\mathfrak{a} \text{ invertible ideal} : [\mathcal{O}_d : \mathfrak{a}] = n\}.$$

Both sets are finite. r_d is the count of Theorem 3.1.2*, R_d that of Corollary 3.1.3*; the thesis prints both with the same display (ERRATA E6.1), we never do.

- (N8) **Standing local setup.** The Chapter 3 counting results assume: $d < 0$ an imaginary quadratic discriminant, $d = Df^2$ with D fundamental, and p a rational prime with $p \mid f$; write $w := v_p(f) \geq 1$ (v_p the p -adic valuation) and $m \in \mathbb{Z}_{\geq 0}$ for the norm exponent. In Lean the d, D, f, p hypotheses are bundled as the class `ConductorPrimeSetup`; w is derived notation (not a field of the bundle) and m is a separate parameter carried by each statement.

- (N9) **The valuation v .** Following ERRATA E6.8/E4, $v_p(d/4)$ is notation:

$$v := v_p(d/4) := \begin{cases} v_p(d), & p \text{ odd}, \\ v_2(d) - 2, & p = 2. \end{cases}$$

Well defined and nonnegative under (N8): for $p = 2$, $2 \mid f$ gives $v_2(d) = v_2(D) + 2w \geq 2$. Explicitly $v = v_p(D) + 2w$ for odd p ($v_p(D) \in \{0, 1\}$) and $v = v_2(D) + 2w - 2$ for $p = 2$ ($v_2(D) \in \{0, 2, 3\}$).

(N10) **Parity flag.** $\varepsilon = \varepsilon(m) := 1$ if m is even, 0 if m is odd; equivalently $\varepsilon(m) = 1 - (m \bmod 2)$. *Warning* (ERRATA E6.6): this is the reverse of the natural parity indicator; the convention is the thesis's own and is kept.

(N11) **Kronecker symbol at p .** For fundamental D ,

$$\chi_p(D) := \begin{cases} \left(\frac{D}{p}\right) \text{ (Legendre)}, & p \text{ odd}, \\ 1, & p = 2, D \equiv 1 \pmod{8}, \\ -1, & p = 2, D \equiv 5 \pmod{8}, \\ 0, & p = 2, 4 \mid D, \end{cases}$$

the cases being exhaustive since odd fundamental D is $\equiv 1 \pmod{4}$. Then p splits in $\mathcal{O}_K$ iff $\chi_p(D) = 1$, is inert iff $\chi_p(D) = -1$, ramifies iff $\chi_p(D) = 0$ iff $p \mid D$. All corrected Chapter 3 statements are phrased p -uniformly through $\chi_p(D)$ (ERRATA E4); there are no separate $p = 2$ displays.

(N12) **Square-root and polynomial-root counts.** For $n \geq 1$ and $c \in \mathbb{Z}$,

$$\text{rc}(c; n) := \#\{x \in \mathbb{Z}/n\mathbb{Z} : x^2 \equiv c \pmod{n}\},$$

the quantity computed by Lemma 3.2.6 for $n = p^s$, $c = p^{2r}u$, $p \nmid u$. Distinct from it is the count of roots of the defining polynomial, $\#\{A \in \mathbb{Z}/N\mathbb{Z} : g(A) \equiv 0 \pmod{N}\}$ ($N \geq 1$), into which the normal-form bijection (Lemma 3.2.4) converts ideal counting and which Lemma 3.2.6 evaluates after completing the square (Lean: `rootCountMod`; the square-root count is `cardSqrts`).

(N13) **Name-collision resolutions** (ERRATA E6.7):

- $R_d(n)$ always denotes the invertible-ideal count (N7); Dorman's maximal-order count (thesis §2.3) is written $R^{\text{Dor}}(n)$ in Chapter 2 documents.
- $\varepsilon(m)$ always denotes the parity flag (N10); the Gross–Zagier character is ϵ_{GZ} and the Lauter–Viray weight $\tilde{\epsilon}$ in Chapter 2 documents.

- s is reserved for the modulus exponent of Lemma 3.2.6; the integer shift of Lemma 3.2.7 is renamed t (that lemma concerns $p^r(\tau - t)$).

Well-definedness: $\frac{d^2-d}{4} \in \mathbb{Z}$ iff $d \equiv 0, 1 \pmod{4}$; uniqueness of (D, f) is standard (take $D = \text{disc } K$, $f = [\mathcal{O}_K : \mathcal{O}_d]$); a nonzero ideal $\mathfrak{a}$ contains $N(\alpha) = \bar{\alpha}\alpha > 0$ for any $0 \neq \alpha \in \mathfrak{a}$, hence $\mathfrak{a} \supseteq N(\alpha)\mathcal{O}_d$ has finite index; ideals of index n are among the finitely many index- n subgroups of $\mathbb{Z}^2$.

Remark 1.2 (Divergence from the thesis).

Uses: §1.1

(i) The symbols $\chi_p(D)$ and the $p = 2$ branch of $v = v_p(d/4)$ do not appear in thesis §1.2; they are introduced so the *corrected* Theorem 3.1.2* and Corollary 3.1.3* (ERRATA E1–E5) can be stated once, p -uniformly, instead of via the thesis’s partly false five-case $p = 2$ display. (ii) ε is promoted from an inline definition in Thm 3.1.2 to standing notation named **epsilonEven**, defusing the reversed-parity trap (E6.6) and the triple ϵ -collision (E6.7). (iii) Invertibility is *defined* by the product-principal condition (N6); the thesis’s “locally principal” description becomes a proved equivalence. (iv) The norm of an ideal is *defined* as the index (N5); the thesis uses $N(\mathfrak{a})$ and $[\mathcal{O} : \mathfrak{a}]$ interchangeably without comment.

Remark 1.3 (Index multiplicativity fails for non-invertible ideals).

Uses: §1.1

With $N(\mathfrak{a}) = [\mathcal{O}_d : \mathfrak{a}]$, $N(\mathfrak{a}\mathfrak{b}) = N(\mathfrak{a})N(\mathfrak{b})$ holds when one factor is invertible or when $\mathfrak{a} + \mathfrak{b} = \mathcal{O}_d$, but not in general: for $d = -12$ ($D = -3$, $f = 2$, $\mathcal{O}_{-12} = \mathbb{Z}[\sqrt{-3}]$) and $\mathfrak{p} = (2, 1 + \sqrt{-3})$ one has $\mathfrak{p}^2 = 2\mathfrak{p}$, so $[\mathcal{O} : \mathfrak{p}^2] = 8 \neq 4 = [\mathcal{O} : \mathfrak{p}]^2$, and $\mathfrak{p}$ is not invertible. Consequently the Lean index layer defines the index as the raw quotient cardinality (`Nat.card`, Mathlib’s `Submodule.cardQuot` convention), not through Mathlib’s Dedekind-gated `Ideal.absNorm` API.

Chapter 2

Infrastructure

The general-purpose layer beneath the thesis’s counting arguments: the index layer replacing Mathlib’s Dedekind-gated norm API, the invertibility interface, and the unit groups of imaginary quadratic orders.

Definition 2.1 (Ideal index).

Lean: `QuadraticOrder.idealIndex`

Uses: §1.1

Let $d < 0$ be an integer with $d \equiv 0$ or $1 \pmod{4}$ and let $\mathcal{O} = \mathcal{O}_d = \mathbb{Z}[\tau]$, $\tau = \frac{d+\sqrt{d}}{2}$, be the imaginary quadratic order of discriminant d , a free $\mathbb{Z}$ -module with basis $\{1, \tau\}$. For an ideal $\mathfrak{a} \subseteq \mathcal{O}$, the *index* (or *norm*) of $\mathfrak{a}$ is

$$[\mathcal{O} : \mathfrak{a}] := \#(\mathcal{O}/\mathfrak{a}) \in \mathbb{Z}_{\geq 0},$$

the cardinality of the additive quotient group when finite and 0 when infinite (the `Nat.card` convention); equivalently, the index of $\mathfrak{a}$ as an additive subgroup of $(\mathcal{O}, +) \cong \mathbb{Z}^2$. The thesis writes $N(\mathfrak{a})$ for this quantity, and “ $\mathfrak{a}$ has norm p^m ” means $[\mathcal{O} : \mathfrak{a}] = p^m$.

Lemma 2.2 (Basic facts about the index).

Uses: §2.1

Let $\mathfrak{a} \subseteq \mathcal{O}$ be an ideal.

1. *For an integer $c \geq 1$, $[\mathcal{O} : c\mathcal{O}] = c^2$.*
2. *If $\mathfrak{a} \neq 0$ then $[\mathcal{O} : \mathfrak{a}] \geq 1$, i.e. the quotient is finite.*
3. *(Lagrange) If $[\mathcal{O} : \mathfrak{a}] = n \geq 1$ then $n \in \mathfrak{a}$, hence $n\mathcal{O} \subseteq \mathfrak{a}$.*
4. *$[\mathcal{O} : \mathfrak{a}] = 1$ if and only if $\mathfrak{a} = \mathcal{O}$.*

Proof.

Uses: §2.1, §1.1

(1) In the $\mathbb{Z}$ -basis $\{1, \tau\}$ we have $\mathcal{O}/c\mathcal{O} \cong (\mathbb{Z}/c)^2$, of order c^2 .

(2) Pick $\alpha \in \mathfrak{a}$, $\alpha \neq 0$. Then $N(\alpha) = \alpha\bar{\alpha} \in \mathfrak{a} \cap \mathbb{Z}$ (it lies in $\mathfrak{a}$ because $\mathfrak{a}$ is an ideal and $\bar{\alpha} \in \mathcal{O}$), and $N(\alpha) > 0$ because the norm form $N(a + b\tau) = a^2 + dab + \frac{d^2-d}{4}b^2$ has discriminant $d < 0$ and is positive definite. With $c := N(\alpha)$ we get $c\mathcal{O} \subseteq \mathfrak{a}$, so $\mathcal{O}/\mathfrak{a}$ is a quotient group of the finite group $\mathcal{O}/c\mathcal{O}$ and is finite, of order dividing c^2 .

(3) The additive group $\mathcal{O}/\mathfrak{a}$ has order n , so by Lagrange's theorem $n \cdot \bar{1} = \bar{0}$, i.e. $n \in \mathfrak{a}$; since $\mathfrak{a}$ is an ideal, $n\mathcal{O} \subseteq \mathfrak{a}$.

(4) An additive subgroup has index 1 exactly when it is the whole group. $\square$

Lemma 2.3 (Finiteness of the set of ideals of index n).

Lean: `QuadraticOrder.finite_setOf_idealIndex_eq`

✓ *Lean-verified*

Uses: §2.1

For every integer $n \geq 1$, the set

$$\{\mathfrak{a} \subseteq \mathcal{O} \text{ ideal} : [\mathcal{O} : \mathfrak{a}] = n\}$$

is finite.

Proof.

Uses: §2.2

Let $\mathfrak{a}$ have index n . By Lemma 2.2(3), $n\mathcal{O} \subseteq \mathfrak{a} \subseteq \mathcal{O}$, so $\mathfrak{a}$ is the preimage under $\pi : \mathcal{O} \rightarrow \mathcal{O}/n\mathcal{O}$ of its image $\mathfrak{a}/n\mathcal{O}$. Hence $\mathfrak{a} \mapsto \mathfrak{a}/n\mathcal{O}$ is injective on the set in question, with target the set of subsets of $\mathcal{O}/n\mathcal{O} \cong (\mathbb{Z}/n)^2$ (Lemma 2.2(1)), a finite set of size 2^{n^2} . Only the fact that $(\mathcal{O}, +)$ is a finitely generated abelian group is used: equivalently, an ideal of index n is a sublattice of $\mathbb{Z}^2$ of index n containing $n\mathcal{O}$, and there are finitely many such (by Hermite normal form, exactly $\sigma(n) = \sum_{t|n} t$ subgroups of $\mathbb{Z}^2$ have index n). $\square$

Proposition 2.4 (Comaximal multiplicativity of the index).

Lean: `QuadraticOrder.idealIndex_mul_of_codisjoint`

✓ *Lean-verified*

Uses: §2.1

Let $I, J \subseteq \mathcal{O}$ be ideals with $I + J = \mathcal{O}$ (in Lean, the hypothesis $I \sqcup J = \top$).

Then

$$[\mathcal{O} : IJ] = [\mathcal{O} : I] \cdot [\mathcal{O} : J].$$

No finiteness hypothesis is needed under the `Nat.card` convention.

Proof.

Uses: §2.1

Step 1: $IJ = I \cap J$. The inclusion $IJ \subseteq I \cap J$ holds for any two ideals. Conversely, by comaximality,

$$I \cap J = (I \cap J) \mathcal{O} = (I \cap J)(I + J) = (I \cap J)I + (I \cap J)J \subseteq JI + IJ = IJ.$$

Step 2: Chinese Remainder. The ring homomorphism $\varphi : \mathcal{O} \rightarrow \mathcal{O}/I \times \mathcal{O}/J$, $x \mapsto (x \bmod I, x \bmod J)$, has kernel $I \cap J$ and is surjective: writing $1 = e + h$ with $e \in I$, $h \in J$, the element $x := ah + be$ satisfies $x \equiv a(1 - e) \equiv a \pmod{I}$ and $x \equiv b(1 - h) \equiv b \pmod{J}$. Hence $\mathcal{O}/(I \cap J) \cong \mathcal{O}/I \times \mathcal{O}/J$.

Step 3. Taking cardinalities, $[\mathcal{O} : IJ] = [\mathcal{O} : I \cap J] = \#(\mathcal{O}/I \times \mathcal{O}/J) = [\mathcal{O} : I][\mathcal{O} : J]$; the last equality holds unconditionally for `Nat.card` since quotient rings are nonempty. The whole argument is valid in any commutative ring. $\square$

Remark 2.5 (Warning: multiplicativity fails without comaximality).

Uses: §2.4

For non-invertible ideals of a non-maximal order the index is *not* multiplicative. In $\mathcal{O}_{-12} = \mathbb{Z}[\sqrt{-3}]$ (here $D = -3$, $f = 2$, $\tau = -6 + \sqrt{-3}$), the unique prime over 2 is $\mathfrak{P} = (2, 1 + \sqrt{-3})$ with $[\mathcal{O} : \mathfrak{P}] = 2$, and since $(1 + \sqrt{-3})^2 = 2(1 + \sqrt{-3}) - 4$ one computes $\mathfrak{P}^2 = (4, 2 + 2\sqrt{-3}) = 2\mathfrak{P}$, whence

$$[\mathcal{O} : \mathfrak{P}^2] = 8 \neq 4 = [\mathcal{O} : \mathfrak{P}]^2.$$

($\mathfrak{P}^2 = 2\mathfrak{P}$ with $\mathfrak{P} \neq 2\mathcal{O}$ also shows $\mathfrak{P}$ is not invertible.) This is why Mathlib's bundled `Ideal.absNorm` (a `MonoidWithZeroHom` from ideals to $\mathbb{N}$) cannot be used for $\mathcal{O}_d$ with $f > 1$: multiplicativity is baked into the monoid-homomorphism type, and its proof (`Submodule.cardQuot_mul`) requires `[IsDedekindDomain S]`, which fails because $\mathcal{O}_d$ is not integrally closed. The failure is mathematical, not an API gap: no multiplicative extension of the index to all ideals exists. The index layer is therefore built directly on `Submodule.cardQuot / AddSubgroup.index / Nat.card`.

Lemma 2.6 (Comaximal families).

Uses: §2.1

Let $I_1, \dots, I_k \subseteq \mathcal{O}$ be pairwise comaximal ideals ($I_i + I_j = \mathcal{O}$ for $i \neq j$). Then

1. $I_1 \cdots I_k = I_1 \cap \cdots \cap I_k$;
2. the natural map $\mathcal{O}/(I_1 \cdots I_k) \rightarrow \prod_{j=1}^k \mathcal{O}/I_j$ is a ring isomorphism;
3. $[\mathcal{O} : I_1 \cdots I_k] = \prod_{j=1}^k [\mathcal{O} : I_j]$.

Proof.

Uses: §2.4

First, if $I + J_1 = \mathcal{O}$ and $I + J_2 = \mathcal{O}$ then

$$\mathcal{O} = (I + J_1)(I + J_2) = I^2 + IJ_2 + J_1I + J_1J_2 \subseteq I + J_1J_2,$$

so I is comaximal with J_1J_2 . Now induct on k , the case $k = 1$ being trivial. Put $P := I_1 \cdots I_{k-1}$; by the sub-claim iterated, $I_k + P = \mathcal{O}$. Then (1) follows from Step 1 of the proof of Proposition 2.4 applied to (P, I_k) together with the inductive hypothesis; (2) from Step 2 applied to (P, I_k) composed with the inductive isomorphism for P ; and (3) from (2) by taking cardinalities. $\square$

Theorem 2.7 (Prime-power factorization of the ideal count).

Lean: *QuadraticOrder.idealCount_multiplicative*

✓ *Lean-verified*

Uses: §2.1

Let $n \geq 1$ and write $m_p := v_p(n)$ for each prime $p \mid n$. Then the map

$$\mathfrak{a} \mapsto (\mathfrak{a} + p^{m_p}\mathcal{O})_{p \mid n}$$

is a bijection

$$\{\mathfrak{a} \subseteq \mathcal{O} : [\mathcal{O} : \mathfrak{a}] = n\} \xrightarrow{\sim} \prod_{p \mid n} \{\mathfrak{b} \subseteq \mathcal{O} : [\mathcal{O} : \mathfrak{b}] = p^{m_p}\},$$

with inverse $(\mathfrak{b}_p)_p \mapsto \prod_p \mathfrak{b}_p = \bigcap_p \mathfrak{b}_p$. Consequently

$$\#\{\mathfrak{a} : [\mathcal{O} : \mathfrak{a}] = n\} = \prod_{p \mid n} \#\{\mathfrak{a} : [\mathcal{O} : \mathfrak{a}] = p^{v_p(n)}\},$$

and the counting function $r(n) := \#\{\mathfrak{a} : [\mathcal{O} : \mathfrak{a}] = n\}$ is multiplicative: $r(n_1 n_2) = r(n_1) r(n_2)$ for $\gcd(n_1, n_2) = 1$.

Proof.

Uses: §2.2, §2.4, §2.6, §2.3

Write $P(n)$ for the set of primes dividing n , so $n = \prod_{p \in P(n)} p^{m_p}$ with all $m_p \geq 1$. If $n = 1$ both sides are singletons ($\{\mathcal{O}\}$ on the left by Lemma 2.2(4), the empty product on the right); assume $n > 1$.

Preliminaries. Let $[\mathcal{O} : \mathfrak{a}] = n$ and set $M := \mathcal{O}/\mathfrak{a}$, a finite $\mathcal{O}$ -module of order n , with quotient map $\pi : \mathcal{O} \rightarrow M$. For $p \in P(n)$ let $M_p := \{x \in M : p^j x = 0 \text{ for some } j \geq 0\}$, an $\mathcal{O}$ -submodule (scalar multiplication commutes with the $\mathbb{Z}$ -action).

Fact A: $M = \bigoplus_{p \in P(n)} M_p$ with $\#M_p = p^{m_p}$. Write $n = p^{m_p} n'_p$ with $p \nmid n'_p$. The integers $\{n'_p\}_{p \in P(n)}$ have no common prime divisor (a prime $q \notin P(n)$ divides no n'_p ; a prime $q \in P(n)$ does not divide n'_q), so $\sum_p c_p n'_p = 1$ for some $c_p \in \mathbb{Z}$. For $x \in M$, $x = \sum_p c_p (n'_p x)$ and $p^{m_p} (n'_p x) = nx = 0$ (Lagrange: $\#M = n$ annihilates M), so $n'_p x \in M_p$ and $M = \sum_p M_p$. For directness: every element of M_q has q -power order dividing n , hence order dividing q^{m_q} ; thus $\sum_{q \neq p} M_q$ is killed by n'_p while M_p is killed by some p^j , and $\gcd(p^j, n'_p) = 1$ forces $M_p \cap \sum_{q \neq p} M_q = 0$. Finally each $\#M_p$ is a power of p (by Cauchy's theorem all prime divisors of $\#M_p$ are p), so $n = \prod_p \#M_p$ and unique factorization give $\#M_p = p^{m_p}$.

Fact B: multiplication by p^{m_p} is zero on M_p and bijective on M_q for $q \neq p$; hence $p^{m_p} M = \bigoplus_{q \neq p} M_q$ and $M/p^{m_p} M \cong M_p$. Indeed $\#M_p = p^{m_p}$ annihilates M_p (Lagrange), and on M_q a kernel element has order dividing both p^{m_p} and q^{m_q} , hence is 0; injectivity on a finite group gives bijectivity.

Define $\mathfrak{a}_p := \mathfrak{a} + p^{m_p} \mathcal{O}$ for $p \in P(n)$.

(i) $[\mathcal{O} : \mathfrak{a}_p] = p^{m_p}$. The image of $\mathfrak{a}_p$ in M is $p^{m_p} M$ and $\mathfrak{a} \subseteq \mathfrak{a}_p$, so $\mathcal{O}/\mathfrak{a}_p \cong M/p^{m_p} M \cong M_p$ by Fact B.

(ii) *Pairwise comaximality.* For $p \neq q$, the coprime integers p^{m_p} and q^{m_q} satisfy $up^{m_p} + vq^{m_q} = 1$, so $1 \in \mathfrak{a}_p + \mathfrak{a}_q$.

(iii) *Injectivity.* By Fact B, $\mathfrak{a}_p = \pi^{-1}(\bigoplus_{q \neq p} M_q)$, and $\bigcap_p \bigoplus_{q \neq p} M_q = 0$ componentwise, so $\bigcap_p \mathfrak{a}_p = \pi^{-1}(0) = \mathfrak{a}$: the tuple of parts determines $\mathfrak{a}$. By (ii) and Lemma 2.6(1) also $\prod_p \mathfrak{a}_p = \mathfrak{a}$.

(iv) *Surjectivity and the inverse.* Let $(\mathfrak{b}_p)_{p \in P(n)}$ satisfy $[\mathcal{O} : \mathfrak{b}_p] = p^{m_p}$. By Lemma 2.2(3), $p^{m_p} \in \mathfrak{b}_p$, so as in (ii) the $\mathfrak{b}_p$ are pairwise comaximal. Set $\mathfrak{a} := \bigcap_p \mathfrak{b}_p = \prod_p \mathfrak{b}_p$ (Lemma 2.6(1)); then $[\mathcal{O} : \mathfrak{a}] = \prod_p p^{m_p} = n$ by Lemma 2.6(3). Let $\psi : \mathcal{O}/\mathfrak{a} \xrightarrow{\sim} \prod_q \mathcal{O}/\mathfrak{b}_q$ be the isomorphism of Lemma 2.6(2). For $x \in \mathcal{O}$ the p -component of $\psi(x + \mathfrak{a})$ vanishes iff $x \in \mathfrak{b}_p$; combined with surjectivity of ψ this gives $\psi(\mathfrak{b}_p/\mathfrak{a}) = H_p := 0 \times \prod_{q \neq p} \mathcal{O}/\mathfrak{b}_q$. On the other hand, multiplication by p^{m_p} kills the factor $\mathcal{O}/\mathfrak{b}_p$ (a group of order p^{m_p}) and is bijective on each $\mathcal{O}/\mathfrak{b}_q$, $q \neq p$ (kernel elements have order di-

viding the coprime integers p^{m_p} and q^{m_q} , so $\psi(p^{m_p}(\mathcal{O}/\mathfrak{a})) = p^{m_p} \prod_q \mathcal{O}/\mathfrak{b}_q = H_p$ as well. Hence $p^{m_p}(\mathcal{O}/\mathfrak{a}) = \mathfrak{b}_p/\mathfrak{a}$, and taking preimages under $\pi_{\mathfrak{a}}$ (both sides contain $\mathfrak{a}$), $\mathfrak{a} + p^{m_p}\mathcal{O} = \mathfrak{b}_p$.

(v) *Conclusion.* Both sides of the bijection are finite sets (Lemma 2.3), so the cardinalities multiply as displayed. For coprime n_1, n_2 : every prime of $n_1 n_2$ divides exactly one n_i , with $v_p(n_1 n_2) = v_p(n_i)$; grouping the factors of the product formula for $n_1 n_2$ gives $r(n_1 n_2) = r(n_1)r(n_2)$. For $p \nmid n$ the factor is $\#\{\mathfrak{b} : [\mathcal{O} : \mathfrak{b}] = 1\} = 1$ by Lemma 2.2(4), so the product may be extended over all primes. The argument is valid in any commutative ring. $\square$

Remark 2.8 (p -parts need not be primary ideals).

Uses: §2.7

The decomposition in Theorem 2.7 is a Sylow-style coprime decomposition, *not* Lasker–Noether primary decomposition: a p -part $\mathfrak{a} + p^{v_p(n)}\mathcal{O}$ has p -power index but need not be a primary ideal. For instance, if $p \nmid f$ splits in $\mathcal{O}$, say $p\mathcal{O} = \mathfrak{p}_1 \mathfrak{p}_2$ with distinct primes $\mathfrak{p}_i$, then $\mathfrak{a} = \mathfrak{p}_1 \mathfrak{p}_2$ has index p^2 and radical $\mathfrak{p}_1 \cap \mathfrak{p}_2$, which is not prime, so $\mathfrak{a}$ is not primary. Lemma 2.9 shows that at primes $p \mid f$ the distinction disappears.

Lemma 2.9 (Bridge lemma: p -power index implies $\mathfrak{P}$ -primary at conductor primes (ERRATA E6.4(a))).

Lean: `QuadraticOrder.isPrimary_of_idealIndex_prime_pow`

✓ *Lean-verified*

Uses: §2.1, §3.3

Suppose $p \mid f$. Then $g(x) \equiv (x - A)^2 \pmod{p}$ for some integer A , and $\mathcal{O}$ has a unique prime ideal containing p , namely the maximal ideal $\mathfrak{P} := p\mathcal{O} + (\tau - A)\mathcal{O}$, of index p . Moreover, every ideal $\mathfrak{a} \subseteq \mathcal{O}$ with $[\mathcal{O} : \mathfrak{a}] = p^m$ for some $m \geq 1$ is $\mathfrak{P}$ -primary; that is, $\mathfrak{a}$ is proper, $xy \in \mathfrak{a}$ implies $x \in \mathfrak{a}$ or $y \in \sqrt{\mathfrak{a}}$, and $\sqrt{\mathfrak{a}} = \mathfrak{P}$.

Proof.

Uses: §2.2, §3.3

Step 1: $g \bmod p$ is the square of a linear polynomial. The discriminant of g is $d = Df^2$. If p is odd, then $p \mid f$ gives $v_p(d) = v_p(D) + 2v_p(f) \geq 2$, in particular $p \mid d$; completing the square in $\mathbb{F}_p[x]$ (where 2 is invertible), $g(x) = (x - d \cdot 2^{-1})^2 - d \cdot 4^{-1} \equiv (x - d \cdot 2^{-1})^2 \pmod{p}$, and we take A a lift of $d \cdot 2^{-1} \bmod p$. If $p = 2$, then $2 \mid f$ gives $v_2(d) = v_2(D) + 2v_2(f) \geq 2$, so $4 \mid d$ and the linear coefficient $-d$ of g is even; hence in $\mathbb{F}_2[x]$, with $c := \frac{d^2 - d}{4} \bmod 2$, $g(x) \equiv x^2 + c = (x + c)^2 \pmod{2}$ (using $c^2 = c$ in $\mathbb{F}_2$), and

we take $A := c$. In both cases $x - A$ is the unique monic irreducible factor of $g \bmod p$.

Step 2: unique prime over p . By Proposition 3.2.1 (3.3), the primes of $\mathcal{O} = \mathbb{Z}[x]/(g)$ containing p are exactly $p\mathcal{O} + (g_i(\tau))$ for g_i the monic irreducible factors of $g \bmod p$; by Step 1 there is exactly one, $\mathfrak{P} = p\mathcal{O} + (\tau - A)\mathcal{O}$. Moreover $\mathcal{O}/\mathfrak{P} \cong \mathbb{F}_p[x]/(g \bmod p, x - A) = \mathbb{F}_p[x]/(x - A) \cong \mathbb{F}_p$ (the middle equality since $g \equiv (x - A)^2 \in (x - A)$), so $\mathfrak{P}$ is maximal of index p .

Step 3: $\sqrt{\mathfrak{a}} = \mathfrak{P}$. Since $p^m > 1$, $\mathfrak{a}$ is proper. By Lemma 2.2(3), $p^m \in \mathfrak{a}$. If $\mathfrak{q} \supseteq \mathfrak{a}$ is prime, then $p^m \in \mathfrak{q}$ forces $p \in \mathfrak{q}$, so $\mathfrak{q} = \mathfrak{P}$ by Step 2. The set of primes containing $\mathfrak{a}$ is nonempty ($\mathfrak{a}$ is proper, and $\mathcal{O}$ is Noetherian, being a finitely generated $\mathbb{Z}$ -module). Since the radical of an ideal is the intersection of the primes containing it, $\sqrt{\mathfrak{a}} = \mathfrak{P}$.

Step 4: maximal radical implies primary. Pass to $R := \mathcal{O}/\mathfrak{a}$; its nilradical is $\mathfrak{P}/\mathfrak{a}$, which is maximal by Step 2. Every maximal ideal of R contains the nilradical, so $\mathfrak{P}/\mathfrak{a}$ is the unique maximal ideal and R is local. If $xy \in \mathfrak{a}$ with $y \notin \sqrt{\mathfrak{a}} = \mathfrak{P}$, then $\bar{y}$ lies outside the maximal ideal of R , hence is a unit, and $\bar{x}\bar{y} = 0$ gives $\bar{x} = 0$, i.e. $x \in \mathfrak{a}$. Thus $\mathfrak{a}$ is primary with radical $\mathfrak{P}$ (in Mathlib: `Ideal.isPrimary_of_isMaximal_radical`). $\square$

Throughout, $d < 0$ is a discriminant ($d \equiv 0, 1 \pmod{4}$), written $d = Df^2$ with D the fundamental discriminant of $K := \mathbb{Q}(\sqrt{d})$ and $f \geq 1$ the conductor; $\sqrt{d} := f\sqrt{D}$. The maximal order is $\mathcal{O}_K = \mathbb{Z}[\omega]$, $\omega := \frac{D+\sqrt{D}}{2}$, and $\mathcal{O} = \mathcal{O}_d := \mathbb{Z}[\tau]$, $\tau := \frac{d+\sqrt{d}}{2}$, is the order of discriminant d : a free $\mathbb{Z}$ -module with basis $1, \tau$, with $g(x) = x^2 - dx + \frac{d^2-d}{4}$ the minimal polynomial of τ . Conjugation $\alpha \mapsto \bar{\alpha}$ preserves $\mathcal{O}$ ($\bar{\tau} = d - \tau$) and $N(\alpha) := \alpha\bar{\alpha} \in \mathbb{Z}$. For a nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}$ the index $[\mathcal{O} : \mathfrak{a}] := \#(\mathcal{O}/\mathfrak{a})$ is finite and is the ideal norm of the development. For an integral domain R with fraction field F and a prime $\mathfrak{p} \subseteq R$ we realize the localization inside F , as $R_{\mathfrak{p}} := \{a/s : a \in R, s \in R \setminus \mathfrak{p}\}$, and write $IR_{\mathfrak{p}}$ for the extension of an ideal $I \subseteq R$. A maximal ideal $\mathfrak{m}$ of $\mathcal{O}$ has residue characteristic q when $\mathfrak{m} \cap \mathbb{Z} = q\mathbb{Z}$ (such a q always exists, by the index layer); we say $\mathfrak{m}$ lies over q .

The thesis never isolates the following notion: its §1.2 speaks of “locally principal (invertible) ideals” and the proof of Corollary 3.1.3 in §3.3 counts locally principal ideals while the statement counts invertible ones. This section is the bridge, stated over a general integral domain where possible.

Definition 2.10 (Invertible ideal).

Lean: `QuadraticOrder.IsInvertibleIdeal`

Uses: §1.1

Let R be an integral domain. An ideal $I \subseteq R$ is *invertible* if there exist an ideal $J \subseteq R$ and an element $x \in R$, $x \neq 0$, such that

$$IJ = xR.$$

The Lean predicate `QuadraticOrder.IsInvertibleIdeal` is this definition for $R = \mathcal{O}_d$. No nonzeroness or finiteness hypothesis on I is imposed; both are consequences (Proposition 2.12).

Remark 2.11.

Uses: §2.10

For a nonzero integral ideal I this agrees with the textbook definition through fractional ideals (Neukirch I §12, Cox §7): if $IJ = x\mathcal{O}$ then $I \cdot (x^{-1}J) = \mathcal{O}$; conversely if $I\mathfrak{b} = \mathcal{O}$ for a fractional ideal $\mathfrak{b}$, pick $0 \neq a \in I$ and set $J := a\mathfrak{b} \subseteq I\mathfrak{b} = \mathcal{O}$, so that $IJ = a\mathcal{O}$. The inverse is unique and equals $(R :_F I)$. The definition was frozen in this integral form because Mathlib's fractional-ideal invertibility API is essentially Dedekind-gated, and $\mathcal{O}_d$ is not Dedekind for $f > 1$.

Proposition 2.12 (Arithmetic of invertible ideals).

Uses: §2.10

Let R be an integral domain with fraction field F and let $I, J, A, B \subseteq R$ be ideals.

- (a) *If I is invertible then $I \neq 0$.*
- (b) *Every nonzero principal ideal is invertible; in particular R is.*
- (c) *If I and J are invertible, so is IJ .*
- (d) *If IJ is invertible, so are I and J .*
- (e) *If I is invertible and $IA = IB$ then $A = B$.*
- (f) *If I is invertible then I is finitely generated.*
- (g) *If R is local and I is invertible, then I is principal, generated by an element of I .*

Proof.

Uses: §2.10

(a) If $I = 0$ then $IJ = 0 \neq xR$ for $x \neq 0$. (b) $(aR) \cdot R = aR$. (c) If $IJ_1 = xR$ and $JJ_2 = yR$ then $(IJ)(J_1J_2) = xyR$ and $xy \neq 0$ in the domain R . (d) If $(IJ)L = xR$ then $I(JL) = xR$ and $J(IL) = xR$. (e) With $IJ = xR$: from $IA = IB$, multiplying by J gives $xA = xB$; for $a \in A$

write $xa = xb$, $b \in B$, and cancel x in the domain. Hence $A \subseteq B$, and symmetrically.

For (f) and (g), let $IJ = xR$, $x \neq 0$. Since $x \in IJ$, write $x = \sum_{i=1}^n a_i c_i$ with $a_i \in I$, $c_i \in J$, and set $b_i := c_i/x \in F$. Then

$$\sum_{i=1}^n a_i b_i = 1 \quad \text{and} \quad b_i I = \frac{1}{x} c_i I \subseteq \frac{1}{x} JI = R \quad (1 \leq i \leq n). \quad (*)$$

(f) For $a \in I$: $a = \sum_i a_i(ab_i)$ with $ab_i \in R$ by (*); so $I = (a_1, \dots, a_n)$. (g) Let $\mathfrak{m}$ be the maximal ideal. Each $a_i b_i$ lies in R by (*); were all of them in $\mathfrak{m}$, we would get $1 = \sum_i a_i b_i \in \mathfrak{m}$. So $u := a_{i_0} b_{i_0}$ is a unit for some i_0 , and every $a \in I$ satisfies $a = a_{i_0} \cdot (u^{-1}(b_{i_0} a)) \in a_{i_0} R$ since $b_{i_0} a \in R$. Hence $I = a_{i_0} R$ with $a_{i_0} \in I$. $\square$

Theorem 2.13 (Invertible iff locally principal).

Lean: `QuadraticOrder.isInvertibleIdeal_iff_isLocallyPrincipal`

Uses: §2.10, §1.1, §2.1

Let R be an integral domain and $I \subseteq R$ a nonzero finitely generated ideal. The following are equivalent:

- (i) I is invertible;
- (ii) $IR_{\mathfrak{p}}$ is principal for every prime ideal $\mathfrak{p}$;
- (iii) $IR_{\mathfrak{m}}$ is principal for every maximal ideal $\mathfrak{m}$;
- (iv) $IR_{\mathfrak{m}}$ is principal for every maximal ideal $\mathfrak{m} \supseteq I$;
- (v) $IR_{\mathfrak{m}}$ is an invertible ideal of $R_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$.

The implications (i) $\Rightarrow$ (ii), (iii), (iv), (v) hold without finite generation. Moreover, when these conditions hold, then with $I^{-1} := (R :_F I) = \{\beta \in F : \beta I \subseteq R\}$ one has $II^{-1} = R$, and for every nonzero $a \in I$ the ideal $J := aI^{-1} \subseteq R$ satisfies $IJ = aR$.

In particular, invertibility is a local property, and since $\mathcal{O}_d$ is a Noetherian domain the equivalence applies to every nonzero ideal of $\mathcal{O}_d$; the Lean declaration is this instance.

Proof.

Uses: §2.10, §2.12, §2.1

We use the standard localization facts, all inside F : elements of $IR_{\mathfrak{p}}$ have the form a/s with $a \in I$, $s \notin \mathfrak{p}$ (clear denominators); $\mathfrak{p}R_{\mathfrak{p}} \cap R = \mathfrak{p}$;

$R_{\mathfrak{p}}$ is local with maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$; extension of ideals is multiplicative, $(IJ)R_{\mathfrak{p}} = (IR_{\mathfrak{p}})(JR_{\mathfrak{p}})$; and $I \neq 0$ implies $IR_{\mathfrak{p}} \neq 0$ (a domain injects into its localizations).

(i) $\Rightarrow$ (v): from $IJ = xR$, extension gives $(IR_{\mathfrak{m}})(JR_{\mathfrak{m}}) = xR_{\mathfrak{m}}$ with $x \neq 0$. (v) $\Rightarrow$ (iii): $R_{\mathfrak{m}}$ is a local domain, so Proposition 2.12(g) applies. (i) $\Rightarrow$ (ii): the same two steps verbatim at an arbitrary prime $\mathfrak{p}$. (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv): specialization. (iv) $\Rightarrow$ (iii): if $I \not\subseteq \mathfrak{m}$, any $s \in I \setminus \mathfrak{m}$ is a unit of $R_{\mathfrak{m}}$ lying in $IR_{\mathfrak{m}}$, so $IR_{\mathfrak{m}} = R_{\mathfrak{m}}$ is principal.

(iii) $\Rightarrow$ (i): here $I = (x_1, \dots, x_r) \neq 0$. The set II^{-1} of finite sums $\sum_k \alpha_k \beta_k$ ($\alpha_k \in I$, $\beta_k \in I^{-1}$) is an ideal of R by definition of I^{-1} . Suppose $II^{-1} \neq R$; choose a maximal ideal $\mathfrak{m} \supseteq II^{-1}$. By (iii) and clearing denominators, $IR_{\mathfrak{m}} = aR_{\mathfrak{m}}$ for some $a \in I$, and $a \neq 0$ since $IR_{\mathfrak{m}} \neq 0$. For each generator $x_j \in IR_{\mathfrak{m}}$ write $x_j = ar_j/s_j$ with $r_j \in R$, $s_j \notin \mathfrak{m}$, i.e. $s_j x_j \in aR$. With $s := s_1 \cdots s_r \notin \mathfrak{m}$ we get $sI \subseteq aR$, hence $s/a \in I^{-1}$ and $s = a \cdot (s/a) \in II^{-1} \subseteq \mathfrak{m}$, a contradiction. Therefore $II^{-1} = R$. Finally, for any nonzero $a \in I$ put $J := aI^{-1}$; then $J \subseteq II^{-1} = R$ is an integral ideal and $IJ = a(II^{-1}) = aR$, proving (i) and the “moreover” clause. (Under hypothesis (i), finite generation holds by Proposition 2.12(f), so the “moreover” clause is unconditional for invertible ideals.) $\square$

Lemma 2.14 (Triviality away from the index).

Uses: §2.10, §1.1, §2.1

Let $\mathfrak{a} \subseteq \mathcal{O}_d$ be an ideal of finite index $n = [\mathcal{O} : \mathfrak{a}]$ and let $\mathfrak{m}$ be a maximal ideal of residue characteristic q with $q \nmid n$. Then $\mathfrak{a} \not\subseteq \mathfrak{m}$ and $\mathfrak{a}\mathcal{O}_{\mathfrak{m}} = \mathcal{O}_{\mathfrak{m}}$. In particular:

- (a) if $[\mathcal{O} : \mathfrak{a}] = p^m$ for a rational prime p and $\mathfrak{q}$ lies over a rational prime $q \neq p$, then $\mathfrak{a}\mathcal{O}_{\mathfrak{q}} = \mathcal{O}_{\mathfrak{q}}$ is the unit ideal;
- (b) every maximal ideal containing $\mathfrak{a}$ has residue characteristic dividing n ; for $n = p^m$, $m \geq 1$, every maximal ideal containing $\mathfrak{a}$ lies over p .

Proof.

Uses: §2.1

By the index layer (Lagrange), $n \in n\mathcal{O} \subseteq \mathfrak{a}$. If $n \in \mathfrak{m}$ then $n \in \mathfrak{m} \cap \mathbb{Z} = q\mathbb{Z}$, i.e. $q \mid n$, contrary to hypothesis. So $n \in \mathfrak{a} \setminus \mathfrak{m}$ is a unit of the local ring $\mathcal{O}_{\mathfrak{m}}$ lying in $\mathfrak{a}\mathcal{O}_{\mathfrak{m}}$, whence $\mathfrak{a}\mathcal{O}_{\mathfrak{m}} = \mathcal{O}_{\mathfrak{m}}$. (a) is the case $n = p^m$, $q \neq p$; (b) is the contrapositive, with $q \mid p^m$ forcing $q = p$ when $m \geq 1$. $\square$

Corollary 2.15 (Invertibility of a p -power-index ideal is a condition at p).

Uses: §2.13, §2.14, §2.1

Let p be a rational prime and let $\mathfrak{a} \subseteq \mathcal{O}_d$ have index p^m , $m \geq 0$. Then $\mathfrak{a}$ is invertible if and only if $\mathfrak{a}\mathcal{O}_{\mathfrak{p}}$ is principal for every maximal ideal $\mathfrak{p}$ of $\mathcal{O}$ lying over p .

Proof.

Uses: §2.13, §2.14, §2.1

Finite index gives $\mathfrak{a} \neq 0$, and $\mathcal{O}_d$ is Noetherian (free of rank 2 over $\mathbb{Z}$), so Theorem 2.13 applies. $(\Rightarrow)$ is (i) $\Rightarrow$ (iii) restricted to the primes over p . $(\Leftarrow)$ verifies (iv): a maximal ideal containing $\mathfrak{a}$ lies over p by Lemma 2.14(b) (for $m = 0$ there is none), so its localization is principal by hypothesis; now apply (iv) $\Rightarrow$ (i). $\square$

Lemma 2.16 (The conductor ideal).

Uses: §1.1

$f\mathcal{O}_K \subseteq \mathcal{O}_d$, and $f\mathcal{O}_K$ is simultaneously an ideal of $\mathcal{O}_K$ and of $\mathcal{O}_d$; moreover $\mathcal{O}_d = \mathbb{Z} + f\mathcal{O}_K$.

Proof.

Uses: §1.1

One computes $\tau - f\omega = \frac{Df^2 + f\sqrt{D}}{2} - \frac{fD + f\sqrt{D}}{2} = \frac{Df(f-1)}{2} \in \mathbb{Z}$ (as $f(f-1)$ is even), so $f\omega \in \mathcal{O}$ and $f\mathcal{O}_K = f\mathbb{Z} + f\omega\mathbb{Z} \subseteq \mathbb{Z} + \tau\mathbb{Z} = \mathcal{O}$ (also $\tau \in \mathcal{O}_K$, so $\mathcal{O} \subseteq \mathcal{O}_K$). Being an $\mathcal{O}_K$ -ideal, $f\mathcal{O}_K$ is a fortiori closed under multiplication by $\mathcal{O} \subseteq \mathcal{O}_K$, hence an $\mathcal{O}$ -ideal. Finally $\mathcal{O} = \mathbb{Z} + \tau\mathbb{Z} \subseteq \mathbb{Z} + f\mathcal{O}_K \subseteq \mathcal{O}$. (Cf. Cox, Lemma 7.2.) $\square$

Lemma 2.17 (DVR away from the conductor).

Uses: §1.1, §2.1, §2.16

Let $\mathfrak{m}$ be a maximal ideal of $\mathcal{O}_d$ of residue characteristic q with $q \nmid f$. Then, inside K :

- (1) $\mathcal{O}_K \subseteq \mathcal{O}_{\mathfrak{m}}$;
- (2) $\mathfrak{M} := \mathfrak{m}\mathcal{O}_{\mathfrak{m}} \cap \mathcal{O}_K$ is a nonzero prime ideal of $\mathcal{O}_K$ with $\mathfrak{M} \cap \mathcal{O} = \mathfrak{m}$;
- (3) $\mathcal{O}_{\mathfrak{m}} = (\mathcal{O}_K)\mathfrak{M}$;
- (4) $\mathcal{O}_{\mathfrak{m}}$ is a discrete valuation ring; in particular $\mathfrak{a}\mathcal{O}_{\mathfrak{m}}$ is principal for every nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}$.

Proof.

Uses: §2.16, §2.1

(1) $f \notin \mathfrak{m}$, else $f \in \mathfrak{m} \cap \mathbb{Z} = q\mathbb{Z}$; so f is a unit of $\mathcal{O}_{\mathfrak{m}}$. For $\beta \in \mathcal{O}_K$, $f\beta \in f\mathcal{O}_K \subseteq \mathcal{O}$ (Lemma 2.16), hence $\beta = (f\beta)/f \in \mathcal{O}_{\mathfrak{m}}$.

(2) $\mathfrak{m}\mathcal{O}_{\mathfrak{m}}$ is the maximal ideal of the local ring $\mathcal{O}_{\mathfrak{m}}$; its contraction $\mathfrak{M}$ to the subring $\mathcal{O}_K$ is prime, and $\mathfrak{M} \cap \mathcal{O} = \mathfrak{m}\mathcal{O}_{\mathfrak{m}} \cap \mathcal{O} = \mathfrak{m}$ by the standard contraction identity. It is nonzero since $q \in \mathfrak{m} \cap \mathcal{O}_K \subseteq \mathfrak{M}$.

(3) $(\subseteq)$ An element α/s of $\mathcal{O}_{\mathfrak{m}}$ ($\alpha \in \mathcal{O}$, $s \in \mathcal{O} \setminus \mathfrak{m}$) has $s \notin \mathfrak{M}$ — else $s \in \mathfrak{M} \cap \mathcal{O} = \mathfrak{m}$ — so $\alpha/s \in (\mathcal{O}_K)_{\mathfrak{M}}$. $(\supseteq)$ An element β/t ($\beta \in \mathcal{O}_K$, $t \in \mathcal{O}_K \setminus \mathfrak{M}$) has $\beta, t \in \mathcal{O}_{\mathfrak{m}}$ by (1) and $t \notin \mathfrak{m}\mathcal{O}_{\mathfrak{m}}$ by definition of $\mathfrak{M}$; so t is a unit of $\mathcal{O}_{\mathfrak{m}}$ and $\beta/t \in \mathcal{O}_{\mathfrak{m}}$.

(4) $\mathcal{O}_K$ is Dedekind and $\mathfrak{M} \neq 0$ is prime, so $(\mathcal{O}_K)_{\mathfrak{M}}$ is a DVR (Neukirch I §11; Atiyah–Macdonald, Thm. 9.3). Nonzero ideals of a DVR are powers of the maximal ideal, hence principal, and $\mathfrak{a}\mathcal{O}_{\mathfrak{m}} \neq 0$. $\square$

Theorem 2.18 (Index prime to the conductor implies invertible).

Lean: `QuadraticOrder.isInvertibleIdeal_of_coprime_conductor`

Uses: §2.10, §1.1, §2.1

Let $\mathfrak{a} \subseteq \mathcal{O}_d$ be an ideal of finite index $n = [\mathcal{O} : \mathfrak{a}]$ with $\gcd(n, f) = 1$. Then $\mathfrak{a}$ is invertible.

Proof.

Uses: §2.13, §2.14, §2.17, §2.1

Finite index gives $\mathfrak{a} \neq 0$, and $\mathcal{O}_d$ is Noetherian; we verify condition (iv) of Theorem 2.13. Let $\mathfrak{m} \supseteq \mathfrak{a}$ be maximal, of residue characteristic q . By Lemma 2.14(b), $q \mid n$; with $\gcd(n, f) = 1$ this forces $q \nmid f$. By Lemma 2.17(4), $\mathfrak{a}\mathcal{O}_{\mathfrak{m}}$ is principal. Hence (iv) holds and $\mathfrak{a}$ is invertible. $\square$

Remark 2.19.

Uses: §2.13, §2.18, §2.15, §2.17

(1) The hypothesis $\gcd([\mathcal{O} : \mathfrak{a}], f) = 1$ is equivalent to the classical “ $\mathfrak{a}$ is prime to the conductor”, $\mathfrak{a} + f\mathcal{O} = \mathcal{O}$ (Cox, Lemma 7.18): one direction is $1 = an + bf \in \mathfrak{a} + f\mathcal{O}$ via $n\mathcal{O} \subseteq \mathfrak{a}$; conversely, if a prime q divides $\gcd(n, f)$, then $\mathfrak{a} + q\mathcal{O} = \mathcal{O}$ would make multiplication by q bijective on $\mathcal{O}/\mathfrak{a}$, contradicting $q \mid n$ by Cauchy’s theorem; so $\mathfrak{a} + f\mathcal{O} \subseteq \mathfrak{a} + q\mathcal{O} \subsetneq \mathcal{O}$. (2) The converse of Theorem 2.18 fails: for $p \mid f$ the principal ideal $p\mathcal{O}$ has index p^2 and is invertible. The corrected Corollary 3.1.3* (ERRATA E5) counts exactly the invertible ideals of p -power index for $p \nmid f$, and by Corollary 2.15 that count is a purely local computation at the primes over p . (3) At a prime over $p \mid f$ the local ring is *not* a DVR, so Lemma 2.17 genuinely requires $q \nmid f$; this is why the invertible count differs from the total count of Theorem 3.1.2*.

Definition 2.20 (Unit count of an imaginary quadratic order).

Uses: §1.1

Let $d < 0$ with $d \equiv 0, 1 \pmod{4}$ be a discriminant and $\mathcal{O}_d = \mathbb{Z}[\tau]$, $\tau = \frac{d+\sqrt{d}}{2}$, the imaginary quadratic order of discriminant d . The *unit count* of $\mathcal{O}_d$ is

$$w(d) := \#\mathcal{O}_d^\times.$$

(In the thesis this quantity appears as $w_i = \#\mathcal{O}_{d_i}^\times$ in the exponents $8/(w_1 w_2)$ and $4/(w_1 w_2)$ of Chapter 2. It must not be confused with the Chapter 3 valuation $w = v_p(f)$ of the conductor; in every node of this batch $w(d)$ means the unit count.)

Lemma 2.21 (Completed square; positive definiteness of the norm).

Uses: §1.1

Let $d < 0$ with $d \equiv 0, 1 \pmod{4}$, and let $N(x + y\tau) = x^2 + dxy + \frac{d^2-d}{4}y^2$ be the norm form of $\mathcal{O}_d$. Then for all $x, y \in \mathbb{Z}$,

$$4N(x + y\tau) = (2x + dy)^2 - dy^2 = (2x + dy)^2 + |d|y^2.$$

Consequently $N(\alpha) \geq 0$ for every $\alpha \in \mathcal{O}_d$, with $N(\alpha) = 0$ if and only if $\alpha = 0$; in particular $N(\alpha) \geq 1$ for all $\alpha \neq 0$.

Proof.

Uses: §1.1

Since $d \equiv 0, 1 \pmod{4}$, the constant term $q := \frac{d^2-d}{4}$ of the minimal polynomial $g(x) = x^2 - dx + q$ of τ is an integer with $4q = d^2 - d$ exactly. Hence

$$4N(x + y\tau) = 4x^2 + 4dxy + (d^2 - d)y^2 = (4x^2 + 4dxy + d^2y^2) - dy^2 = (2x + dy)^2 - dy^2,$$

and $-d = |d|$ because $d < 0$. Both summands are squares times nonnegative integers, so $4N \geq 0$, i.e. $N \geq 0$. If $N(x + y\tau) = 0$ then $|d|y^2 = 0$ and $(2x + dy)^2 = 0$; since $|d| \geq 3 > 0$, the first equation forces $y = 0$ and then the second forces $x = 0$. As $\{1, \tau\}$ is a $\mathbb{Z}$ -basis of $\mathcal{O}_d$, this means $\alpha = 0$. $\square$

Lemma 2.22 (Units have norm one).

Uses: §1.1, §2.21

With the hypotheses of Lemma 2.21, an element $\alpha \in \mathcal{O}_d$ is a unit if and only if $N(\alpha) = 1$.

Proof.

Uses: §2.21

Write $\bar{\tau} := d - \tau$; from $g(\tau) = 0$ one computes $g(\bar{\tau}) = (d - \tau)^2 - d(d - \tau) + q = \tau^2 - d\tau + q = 0$, so $\bar{\tau}$ is the other root of g and the universal property of $\mathbb{Z}[x]/(g)$ yields a ring homomorphism $\sigma : \mathcal{O}_d \rightarrow \mathcal{O}_d$ with $\sigma(\tau) = \bar{\tau}$, explicitly

$\sigma(x + y\tau) = (x + dy) - y\tau$; it is an involution fixing $\mathbb{Z}$. Vieta gives $\tau + \bar{\tau} = d$ and $\tau\bar{\tau} = q$, whence

$$\alpha \sigma(\alpha) = (x + y\tau)(x + y\bar{\tau}) = x^2 + xy(\tau + \bar{\tau}) + y^2\tau\bar{\tau} = N(\alpha) \cdot 1,$$

and since σ is a ring homomorphism, N is multiplicative: $N(\alpha\beta) = \alpha\beta \sigma(\alpha) \sigma(\beta) = N(\alpha)N(\beta)$, $N(1) = 1$.

If $\alpha\beta = 1$, then $N(\alpha)N(\beta) = 1$ with both factors nonnegative integers by Lemma 2.21 (both $\alpha, \beta \neq 0$), so $N(\alpha) = 1$. Conversely, if $N(\alpha) = 1$ then $\alpha \sigma(\alpha) = 1$ with $\sigma(\alpha) \in \mathcal{O}_d$, so α is a unit. $\square$

Theorem 2.23 (Units of $\mathcal{O}_d$ for $d < -4$).

Lean: `QuadraticOrder.units_eq_one_or_neg_one`

✓ *Lean-verified*

Uses: §1.1

Let $d < -4$ with $d \equiv 0, 1 \pmod{4}$. Then

$$\mathcal{O}_d^\times = \{1, -1\}.$$

Proof.

Uses: §2.21, §2.22

Let $\alpha = x + y\tau$ be a unit. By Lemma 2.22 and Lemma 2.21,

$$(2x + dy)^2 + |d|y^2 = 4N(\alpha) = 4.$$

If $y \neq 0$, then $|d|y^2 \geq |d| \geq 5$ (since $d < -4$ means $d \leq -5$), which exceeds 4 while $(2x + dy)^2 \geq 0$ — a contradiction. Hence $y = 0$ and $4x^2 = 4$, so $x = \pm 1$, i.e. $\alpha = \pm 1$. Conversely $N(\pm 1) = 1$, so both are units, and $1 \neq -1$ because $\mathcal{O}_d$ is a free $\mathbb{Z}$ -module of rank 2, hence of characteristic 0. $\square$

Corollary 2.24 ($w(d) = 2$ for $d < -4$).

Lean: `QuadraticOrder.card_units_eq_two`

✓ *Lean-verified*

Uses: §2.23, §2.20

Let $d < -4$ with $d \equiv 0, 1 \pmod{4}$. Then $w(d) = \#\mathcal{O}_d^\times = 2$.

Proof.

Uses: §2.23

Immediate from Theorem 2.23: the unit group is $\{1, -1\}$ and these two elements are distinct in characteristic 0. $\square$

Proposition 2.25 (Exceptional discriminants; the unit-count trichotomy).

Uses: §2.20, §2.21, §2.22, §2.23

For the two remaining negative discriminants:

1. $\mathcal{O}_{-4}^\times = \{\pm 1, \pm(2 + \tau)\} \cong \mathbb{Z}/4$, generated by $2 + \tau$ (which is i under $\mathcal{O}_{-4} \cong \mathbb{Z}[i]$, $\tau \mapsto -2 + i$); so $w(-4) = 4$.
2. $\mathcal{O}_{-3}^\times = \{\pm 1, \pm(1 + \tau), \pm(2 + \tau)\} \cong \mathbb{Z}/6 = \mu_6$, with $1 + \tau \mapsto \zeta_3$ and $2 + \tau \mapsto \zeta_6$ under $\tau \mapsto \frac{-3 + \sqrt{-3}}{2}$; so $w(-3) = 6$.

Consequently, for every negative discriminant d : $w(d) \in \{2, 4, 6\}$; $w(d) = 4$ iff $d = -4$; $w(d) = 6$ iff $d = -3$; and every unit of $\mathcal{O}_d$ is a root of unity, so $w(d)$ also equals the number of roots of unity in $\mathcal{O}_d$ (reconciling the two definitions of w_i used in the thesis, §2.1 vs §2.2).

Proof.

Uses: §2.21, §2.22, §2.23

Case $d = -4$. Here $q = 5$, so $\tau^2 = -4\tau - 5$, and Lemma 2.21 gives $4N = (2x - 4y)^2 + 4y^2$, i.e. $N(x + y\tau) = (x - 2y)^2 + y^2$. By Lemma 2.22, units correspond to $(x - 2y)^2 + y^2 = 1$, i.e. $(x - 2y, y) \in \{(\pm 1, 0), (0, \pm 1)\}$, i.e. $(x, y) \in \{(\pm 1, 0), (2, 1), (-2, -1)\}$: the four elements $\pm 1, \pm(2 + \tau)$. Moreover $(2 + \tau)^2 = 4 + 4\tau + \tau^2 = -1$, so $2 + \tau$ has order 4 and the group is cyclic.

Case $d = -3$. Here $q = 3$, so $\tau^2 = -3\tau - 3$, and units correspond to $(2x - 3y)^2 + 3y^2 = 4$. From $3y^2 \leq 4$ we get $y \in \{0, \pm 1\}$: $y = 0$ gives $x = \pm 1$; $y = 1$ gives $(2x - 3)^2 = 1$, so $x \in \{1, 2\}$; $y = -1$ gives $(2x + 3)^2 = 1$, so $x \in \{-1, -2\}$. That is six units $\pm 1, \pm(1 + \tau), \pm(2 + \tau)$. Moreover $(1 + \tau)^2 = 1 + 2\tau + \tau^2 = -(2 + \tau)$ and $(1 + \tau)^3 = -(1 + \tau)(2 + \tau) = -(2 + 3\tau + \tau^2) = 1$, so $1 + \tau$ has order 3 and $-(1 + \tau)$ has order 6: the group is cyclic of order 6.

Trichotomy. The integers $-1, -2, -5, -6$ are not discriminants ($\equiv 3, 2, 3, 2 \pmod{4}$ respectively), so every negative discriminant other than $-3, -4$ satisfies $d \leq -7 < -4$ and falls under Theorem 2.23. Hence $w(d) = 2$ for $d < -4$, $w(-4) = 4$, $w(-3) = 6$, which is the stated trichotomy.

Torsion. $\mathcal{O}_d^\times$ is a finite group of order $w(d)$, so by Lagrange $u^{w(d)} = 1$ for every unit u ; conversely every root of unity is a unit. Hence the number of units equals the number of roots of unity. $\square$

Chapter 3

Ideal counting in imaginary quadratic orders

Chapter 3 of the thesis, with the corrected statements of `ERRATA.md`: the prime classification, the localization suite, the normal-form keystone, the root-counting engine, and the two headline counting theorems, completed by the glue results for arbitrary norms.

Throughout this section d is a non-square integer with $d \equiv 0, 1 \pmod{4}$, written $d = Df^2$ with D the fundamental discriminant of $K = \mathbb{Q}(\sqrt{d})$ and $f \geq 1$ the conductor; $\tau = (d + \sqrt{d})/2$ has minimal polynomial $g(X) = X^2 - dX + \frac{d^2-d}{4} \in \mathbb{Z}[X]$, and $\mathcal{O} = \mathcal{O}_d = \mathbb{Z}[\tau] = \mathbb{Z} \oplus \mathbb{Z}\tau \cong \mathbb{Z}[X]/(g)$ is the order of discriminant d , with $\mathcal{O}_K = \mathcal{O}_D$ the maximal order. For a prime p , $\bar{g} \in \mathbb{F}_p[X]$ denotes the coefficientwise reduction of g modulo p . Negativity of d is nowhere needed in this section.

Definition 3.1 (Splitting type in a quadratic order).

Uses: §1.1

Let p be a rational prime and $\mathcal{O} = \mathcal{O}_d$. We say

- p is *inert* in $\mathcal{O}$ if $p\mathcal{O}$ is a maximal ideal of $\mathcal{O}$;
- p is *split* in $\mathcal{O}$ if $p\mathcal{O}$ is a radical ideal of $\mathcal{O}$ but not maximal;
- p is *ramified* in $\mathcal{O}$ if $p\mathcal{O}$ is not a radical ideal of $\mathcal{O}$.

Since maximal ideals are prime and prime ideals are radical, exactly one of the three alternatives holds for each p . Because $\mathcal{O}$ is not a Dedekind domain when $f > 1$, this ideal-theoretic definition replaces the classical one via factorization of $p\mathcal{O}$ into primes; Theorem 3.4 shows the two agree whenever the classical one applies (in particular in $\mathcal{O}_K$).

Lemma 3.2 (Quotient presentation of $\mathcal{O}/p\mathcal{O}$).

Lean: *QuadraticOrder.quadraticOrderModP_equiv_polyModQuot*

Uses: §1.1

For every rational prime p there is a ring isomorphism

$$\Phi: \mathcal{O}/p\mathcal{O} \xrightarrow{\sim} \mathbb{F}_p[X]/(\bar{g}), \quad \Phi(\tau \bmod p\mathcal{O}) = X \bmod (\bar{g}), \quad \Phi(n \bmod p\mathcal{O}) = \bar{n} \quad (n \in \mathbb{Z}).$$

Proof.

Uses: §1.1

Identify $\mathcal{O} = \mathbb{Z}[X]/(g)$. Let $\pi: \mathbb{Z}[X] \rightarrow \mathbb{F}_p[X]$ be coefficientwise reduction, a surjective ring homomorphism with kernel $p\mathbb{Z}[X]$.

Step 1: $\mathbb{Z}[X]/(p, g) \cong \mathbb{F}_p[X]/(\bar{g})$. The composite $\psi: \mathbb{Z}[X] \rightarrow \mathbb{F}_p[X]/(\bar{g})$ is surjective with kernel $\pi^{-1}((\bar{g})) = p\mathbb{Z}[X] + g\mathbb{Z}[X]$: the inclusion $\supseteq$ holds since $\pi(p) = 0$ and $\pi(g) = \bar{g}$; conversely if $\pi(h) = \bar{g}\bar{k}$, lift $\bar{k}$ to $k \in \mathbb{Z}[X]$, so $\pi(h - gk) = 0$ and $h \in p\mathbb{Z}[X] + g\mathbb{Z}[X]$. Apply the first isomorphism theorem.

Step 2: $\mathcal{O}/p\mathcal{O} \cong \mathbb{Z}[X]/(p, g)$. The preimage of $p\mathcal{O}$ under the quotient map $\rho: \mathbb{Z}[X] \rightarrow \mathbb{Z}[X]/(g) = \mathcal{O}$ is $(p) + (g)$: if $h(\tau) = p\beta$ with $\beta \in \mathcal{O}$, lift β to $b \in \mathbb{Z}[X]$; then $h - pb \in \ker \rho = (g)$. Apply the third isomorphism theorem.

Composing the two isomorphisms yields Φ ; chasing X and the constants gives the stated values. (Primality of p is not used; the Lean statement assumes it only because that is its sole use site.) $\square$

Proposition 3.3 (Primes of $\mathbb{Z}[\tau]$ above p — thesis Proposition 3.2.1).

Uses: §1.1

Let p be a rational prime and let $\bar{g} = \prod_{i=1}^r \bar{g}_i^{e_i}$ be the factorization of $\bar{g}$ into powers of pairwise distinct monic irreducible polynomials $\bar{g}_i \in \mathbb{F}_p[X]$. For each i choose a lift $g_i \in \mathbb{Z}[X]$ of $\bar{g}_i$. Then the prime ideals of $\mathcal{O}$ containing p are exactly

$$\mathfrak{p}_i = p\mathcal{O} + g_i(\tau)\mathcal{O}, \quad i = 1, \dots, r,$$

pairwise distinct, independent of the choice of lifts, and all maximal, with residue fields $\mathcal{O}/\mathfrak{p}_i \cong \mathbb{F}_p[X]/(\bar{g}_i) \cong \mathbb{F}_{p^{\deg \bar{g}_i}}$. In particular the number of primes of $\mathcal{O}$ above p equals the number $r \in \{1, 2\}$ of distinct monic irreducible factors of $\bar{g}$.

Proof.

Uses: §3.2

We chain three inclusion-preserving bijections.

(a) By the correspondence theorem for $\mathcal{O} \twoheadrightarrow \mathcal{O}/p\mathcal{O}$, primes of $\mathcal{O}$ containing p (equivalently, containing $p\mathcal{O}$) correspond to primes of $\mathcal{O}/p\mathcal{O}$, preserving inclusions, maximality, and residue rings.

(b) By Lemma 3.2, these correspond to primes of $\mathbb{F}_p[X]/(\bar{g})$, which by the correspondence theorem again are the primes of $\mathbb{F}_p[X]$ containing $\bar{g}$.

(c) $\mathbb{F}_p[X]$ is a PID: its primes are (0) and (π) with π irreducible, and each nonzero prime has a unique monic generator. Since $\bar{g}$ is monic, (0) does not contain it, and $\bar{g} \in (\pi) \iff \pi \mid \bar{g} \iff \pi \sim \bar{g}_i$ for some i , by unique factorization. So the primes of $\mathbb{F}_p[X]$ containing $\bar{g}$ are exactly the pairwise distinct $(\bar{g}_1), \dots, (\bar{g}_r)$.

Explicit pullback. Let $\chi: \mathcal{O} \rightarrow \mathbb{F}_p[X]/(\bar{g})$ be the composite; for $\alpha = h(\tau)$ one has $\chi(\alpha) = \bar{h} \bmod (\bar{g})$. Fix i and let $\mathfrak{p}_i$ be the pullback of $(\bar{g}_i)/(\bar{g})$. Then $p, g_i(\tau) \in \mathfrak{p}_i$, so $p\mathcal{O} + g_i(\tau)\mathcal{O} \subseteq \mathfrak{p}_i$. Conversely if $\alpha = h(\tau) \in \mathfrak{p}_i$, then $\bar{h} \equiv \bar{k}\bar{g}_i \pmod{\bar{g}}$; lifting $\bar{k}$ to k and the cofactor to l gives $h - kg_i - lg \in p\mathbb{Z}[X]$, and evaluating at τ (where $g(\tau) = 0$) gives $\alpha \in p\mathcal{O} + g_i(\tau)\mathcal{O}$. Two lifts of $\bar{g}_i$ differ by an element of $p\mathbb{Z}[X]$, so $\mathfrak{p}_i$ is independent of the lift. Finally

$$\mathcal{O}/\mathfrak{p}_i \cong (\mathbb{F}_p[X]/(\bar{g})) / ((\bar{g}_i)/(\bar{g})) \cong \mathbb{F}_p[X]/(\bar{g}_i),$$

a field with $p^{\deg \bar{g}_i}$ elements, so each $\mathfrak{p}_i$ is maximal. Since $\deg \bar{g} = 2$, the factorization is an irreducible quadratic, two distinct monic linear factors, or the square of a monic linear factor; hence $r \in \{1, 2\}$. $\square$

Theorem 3.4 (Trichotomy: inert / split / ramified).

Lean: `QuadraticOrder.prime_inert_iff, QuadraticOrder.prime_ramified_iff, QuadraticOrder.prime_split_iff, QuadraticOrder.polyMod_discrim_eq, QuadraticOrder.polyMod_eq_X_sq_of_p_dvd_d, QuadraticOrder.quadraticOrderModP_equiv_X_sq_quot`

Uses: §1.1, §3.1

Let p be a rational prime. Exactly one of the following holds, and within each case all listed conditions are equivalent:

1. **(inert)** $\bar{g}$ is irreducible in $\mathbb{F}_p[X] \iff p\mathcal{O}$ is maximal. Then $\mathcal{O}/p\mathcal{O} \cong \mathbb{F}_{p^2}$ and $p\mathcal{O}$ itself is the unique prime of $\mathcal{O}$ containing p .
2. **(split)** $\bar{g} = (X - \bar{r})(X - \bar{s})$ with $\bar{r} \neq \bar{s} \iff p\mathcal{O}$ is radical but not maximal. Then $\mathcal{O}/p\mathcal{O} \cong \mathbb{F}_p \times \mathbb{F}_p$ and the primes above p are exactly $p\mathcal{O} + (\tau - r)\mathcal{O}$ and $p\mathcal{O} + (\tau - s)\mathcal{O}$ for integer lifts r, s .
3. **(ramified)** $\bar{g} = (X - \bar{A})^2 \iff p\mathcal{O}$ is not radical. Then $\mathcal{O}/p\mathcal{O} \cong \mathbb{F}_p[Y]/(Y^2)$ and $\mathfrak{p} = p\mathcal{O} + (\tau - A)\mathcal{O}$ is the unique prime above p (any integer lift A), with $\mathfrak{p}^2 \subseteq p\mathcal{O} \subsetneq \mathfrak{p}$.

Moreover the cases are detected by symbols and congruences: for odd p , case

$$(1) \iff \left(\frac{d}{p}\right) = -1, \text{ case (2)} \iff \left(\frac{d}{p}\right) = 1, \text{ case (3)} \iff \left(\frac{d}{p}\right) =$$

$0 \iff p \mid d$; for $p = 2$, case (1) $\iff d \equiv 5 \pmod{8}$, case (2) $\iff d \equiv 1 \pmod{8}$, case (3) $\iff 2 \mid d$ (equivalently $4 \mid d$).

Proof.

Uses: §3.3, §3.2

Step 0 (shapes). A monic quadratic over $\mathbb{F}_p$ either has no root (hence is irreducible) or factors into two monic linear factors, distinct or equal: exactly one of the three shapes holds.

Step 1 (shape $\Rightarrow$ ideal property). (1) If $\bar{g}$ is irreducible, $(\bar{g})$ is maximal in the PID $\mathbb{F}_p[X]$, so $\mathbb{F}_p[X]/(\bar{g})$ is a field with p^2 elements, i.e. $\mathbb{F}_{p^2}$; by Lemma 3.2, $p\mathcal{O}$ is maximal. By Proposition 3.3 with $r = 1$ and lift $g_1 = g$, the unique prime above p is $p\mathcal{O} + g(\tau)\mathcal{O} = p\mathcal{O}$, since $g(\tau) = 0$. (2) If $\bar{g} = (X - \bar{r})(X - \bar{s})$ with $\bar{r} \neq \bar{s}$, the factors generate comaximal ideals (their difference $\bar{s} - \bar{r}$ is a unit), so by the Chinese remainder theorem $\mathbb{F}_p[X]/(\bar{g}) \cong \mathbb{F}_p \times \mathbb{F}_p$. A product of fields is reduced, and an ideal is radical iff its quotient is reduced; transporting along Φ , $p\mathcal{O}$ is radical. It is not maximal since $\mathbb{F}_p \times \mathbb{F}_p$ is not a domain. Proposition 3.3 with lifts $X - r$, $X - s$ gives the two primes. (3) If $\bar{g} = (X - \bar{A})^2$, comparing coefficients gives $2A \equiv d$ and $A^2 \equiv \frac{d^2 - d}{4} \pmod{p}$, whence $g(A) \equiv A^2 - 2A \cdot A + A^2 = 0 \pmod{p}$. The identity $g(A + Y) = Y^2 + (2A - d)Y + g(A)$ evaluated at $Y = \tau - A$, together with $g(\tau) = 0$, gives

$$(\tau - A)^2 = (d - 2A)(\tau - A) - g(A) \in p\mathcal{O},$$

since $p \mid d - 2A$ and $p \mid g(A)$. But $\tau - A \notin p\mathcal{O} = p\mathbb{Z} \oplus p\mathbb{Z}\tau$, as its τ -coordinate is 1. So $p\mathcal{O}$ is not radical, with witness $\tau - A$. Translating $X \mapsto X + \bar{A}$ shows $\mathcal{O}/p\mathcal{O} \cong \mathbb{F}_p[Y]/(Y^2)$. Proposition 3.3 ($r = 1$, lift $X - A$) gives the unique prime $\mathfrak{p} = p\mathcal{O} + (\tau - A)\mathcal{O}$; its square is generated by p^2 , $p(\tau - A)$, $(\tau - A)^2$, all in $p\mathcal{O}$, and $\tau - A \in \mathfrak{p} \setminus p\mathcal{O}$ gives $p\mathcal{O} \subsetneq \mathfrak{p}$.

Step 2 (ideal property $\Rightarrow$ shape). The three shapes partition all possibilities (Step 0), and so do the three ideal properties (radical or not; if radical, maximal or not; maximal $\Rightarrow$ radical). Since shape i implies property i for each i , disjointness forces the converses.

Step 3 (symbols, p odd). The constant term of $\bar{g}$ is $(\bar{d}^2 - \bar{d})\bar{4}^{-1}$ (as $4 \cdot \frac{d^2 - d}{4} = d^2 - d$ in $\mathbb{Z}$), so completing the square — $\bar{g} = (X - \bar{d}/2)^2 - \bar{d}/4$ — and applying the automorphism $X \mapsto X + \bar{d}/2$ shows $\bar{g}$ has the shape of $Y^2 - \bar{d}/4$. Equivalently, $\text{disc}(\bar{g}) = \bar{d}^2 - 4 \cdot (\bar{d}^2 - \bar{d})/4 = \bar{d}$. Since $\bar{4}$ is a nonzero square, $Y^2 - \bar{d}/4$ is irreducible / has two distinct roots / has a double root according as $\bar{d}$ is a nonsquare / a nonzero square / zero, i.e. $\left(\frac{d}{p}\right) = -1, 1, 0$; and $\left(\frac{d}{p}\right) = 0 \iff p \mid d$.

Step 4 ($p = 2$). Over $\mathbb{F}_2$, $\bar{g} = X^2 + \bar{d}X + \bar{q}$ with $q = \frac{d(d-1)}{4}$. If $d \equiv 0 \pmod{4}$ (the only even case for a discriminant) then $\bar{d} = 0$ and $\bar{g} = X^2 + \bar{q} = (X + \bar{q})^2$ (Frobenius), shape (3). If $d \equiv 1 \pmod{4}$ then $\bar{d} = 1$ and $\bar{q} = (d-1)/4$ (as d is odd): for $d \equiv 1 \pmod{8}$, $\bar{q} = 0$ and $\bar{g} = X(X+1)$, shape (2); for $d \equiv 5 \pmod{8}$, $\bar{q} = 1$ and $\bar{g} = X^2 + X + 1$ has no root in $\mathbb{F}_2$, shape (1). $\square$

Lemma 3.5 (Split criterion — thesis Remark 3.2.3, promoted).

Uses: §1.1, §3.1

For every rational prime p :

$$p \text{ splits in } \mathcal{O}_d \iff p \text{ splits in } \mathcal{O}_K \text{ and } p \nmid f,$$

where splitting in $\mathcal{O}_K = \mathcal{O}_D$ is Definition 3.1 applied with d replaced by D (agreeing, by Theorem 3.4, with the classical notion $p\mathcal{O}_K = \mathfrak{q}\mathfrak{q}'$, $\mathfrak{q} \neq \mathfrak{q}'$).

Proof.

Uses: §3.4

Claim 1: if $p \mid f$ then $\bar{g}_d$ is the square of a monic linear polynomial; in particular p is ramified, hence not split, in $\mathcal{O}_d$. Since $d = Df^2$, $p \mid f$ gives $p^2 \mid d$. For odd p : $\bar{d} = 0$, and from $4 \cdot \frac{d^2-d}{4} = d(d-1) \equiv 0 \pmod{p}$ with $p \nmid 4$ also $(d^2-d)/4 = 0$, so $\bar{g}_d = X^2$. For $p = 2$: $4 \mid d$, and Step 4 of Theorem 3.4 gives $\bar{g}_d = (X + \bar{q})^2$. Either way, shape (3).

Claim 2: if $p \nmid f$ then $\bar{g}_d$ and $\bar{g}_D$ have the same shape, so p has the same splitting type in $\mathcal{O}_d$ and $\mathcal{O}_K$. Set $c := \frac{d-Df}{2} = \frac{Df(f-1)}{2} \in \mathbb{Z}$ ($f(f-1)$ is even). Direct expansion verifies the identity in $\mathbb{Z}[X]$:

$$g_d(fX + c) = f^2 g_D(X) :$$

the linear coefficient of the left side is $f(2c-d) = f(-Df) = f^2(-D)$, and its constant term is $c^2 - dc + \frac{d^2-d}{4} = \frac{(d-Df)^2 - 2d(d-Df) + d^2-d}{4} = \frac{D^2f^2-d}{4} = f^2 \cdot \frac{D^2-D}{4}$. Reducing mod p : $\bar{g}_d(\bar{f}X + \bar{c}) = \bar{f}^2 \bar{g}_D(X)$. Since $\bar{f} \in \mathbb{F}_p^\times$, the substitution $h(X) \mapsto h(\bar{f}X + \bar{c})$ is an $\mathbb{F}_p$ -algebra automorphism of $\mathbb{F}_p[X]$, preserving degrees, irreducibility and multiplicities; and it carries $\bar{g}_d$ to a unit multiple of $\bar{g}_D$. Hence the two reductions have the same shape, and Theorem 3.4 (applied at discriminant d and at discriminant D) transfers the splitting type.

If p splits in $\mathcal{O}_d$, Claim 1 (contrapositive) gives $p \nmid f$ and Claim 2 gives splitting in $\mathcal{O}_K$; the converse is Claim 2 again. (For odd p the symbol short-cut $\left(\frac{d}{p}\right) = \left(\frac{D}{p}\right) \left(\frac{f}{p}\right)^2 = \left(\frac{D}{p}\right)$ replaces Claim 2; the substitution argument is used because it covers $p = 2$ uniformly.) $\square$

Proposition 3.6 (Unique prime above a conductor prime — corollary of Remark 3.2.3).

Lean: `QuadraticOrder.existsUnique_isPrime_mem_of_dvd_conductor`

✓ *Lean-verified*

Uses: §1.1

Suppose $p \mid f$. Then $\bar{g} = (X - \bar{A})^2$ is the square of a monic linear polynomial ($A = 0$ works for odd p , and $A = \frac{d^2-d}{4}$ for $p = 2$), and there is exactly one prime ideal of $\mathcal{O}$ containing p , namely

$$\mathfrak{p} = p\mathcal{O} + (\tau - A)\mathcal{O} = p\mathbb{Z} \oplus (\tau - A)\mathbb{Z}, \quad [\mathcal{O} : \mathfrak{p}] = p, \quad \mathcal{O}/\mathfrak{p} \cong \mathbb{F}_p.$$

This uniqueness is the standing hypothesis of all of §§3.2–3.3 of the thesis (Lemmas 3.2.2, 3.2.4, 3.2.7, 3.2.8), which is unrestated there (ERRATA E6.2).

Proof.

Uses: §3.3, §3.5, §3.4

By Claim 1 of Lemma 3.5, $\bar{g} = (X - \bar{A})^2$ with the stated values of A (for $p = 2$ note $(X + \bar{q})^2 = (X - \bar{q})^2$ over $\mathbb{F}_2$). So $\bar{g}$ has exactly one distinct monic irreducible factor $X - \bar{A}$, and Proposition 3.3 gives exactly one prime above p , namely $\mathfrak{p} = p\mathcal{O} + (\tau - A)\mathcal{O}$, with $\mathcal{O}/\mathfrak{p} \cong \mathbb{F}_p[X]/(X - \bar{A}) \cong \mathbb{F}_p$ and hence $[\mathcal{O} : \mathfrak{p}] = p$.

For the $\mathbb{Z}$ -module form, $\mathfrak{p}$ is generated as a $\mathbb{Z}$ -module by $p, p\tau, \tau - A, (\tau - A)\tau$. Now $p\tau = p(\tau - A) + pA$, and by the identity of Theorem 3.4 (Step 1, case 3), $(\tau - A)\tau = (d - A)(\tau - A) - g(A)$ with $g(A) \in p\mathbb{Z}$. So $\mathfrak{p} = p\mathbb{Z} + (\tau - A)\mathbb{Z}$, and the sum is direct because p and $\tau - A$ have coordinates $(p, 0)$ and $(-A, 1)$ in the basis $\{1, \tau\}$. $\square$

Remark (divergence from the thesis: proof route). The thesis proves Proposition 3.2.1 by writing a candidate prime as $\mathfrak{p} = (p) + I$ and asserting that, for $\mathcal{O}/\mathfrak{p} \cong (\mathbb{Z}/p[x]/(\prod_i \bar{g}_i))/\bar{I}$ to be a domain, $\bar{I}$ must contain some $(\bar{g}_i)$. Following the existing Lean formalization, this blueprint instead routes everything through the single isomorphism Φ of Lemma 3.2 and transports the ideal-theoretic properties (maximality, radicality, reducedness of the quotient) across it, letting the PID structure of $\mathbb{F}_p[X]$ do the work. This makes explicit what the thesis leaves implicit: the classification of primes of $\mathbb{F}_p[X]/(\bar{g})$ behind the "domain" step, the pairwise distinctness of the $\mathfrak{p}_i$, the independence of the lifts g_i , and the maximality of every prime above p .

Remark (Remark 3.2.3 was unproved and is load-bearing). In the thesis, Remark 3.2.3 is asserted without proof, yet the standing hypothesis of §§3.2–3.3 (“ $p \mid f$, hence a unique prime above p ”) depends on it; ERRATA E6.3 flags this and sketches the proof adopted in Claim 1 of Lemma 3.5. Proposition 3.6 promotes the uniqueness consequence to a standalone statement.

Remark (ramified at a conductor prime is not $p\mathcal{O} = \mathfrak{p}^2$). For $p \mid f$ only $\mathfrak{p}^2 \subseteq p\mathcal{O} \subsetneq \mathfrak{p}$ survives of the Dedekind picture. For $d = -27$ ($D = -3$, $f = 3$, $p = 3$) one computes $\mathfrak{p} = (3, \tau)$ and $\mathfrak{p}^2 = (9, 3\tau) = 3\mathfrak{p}$ of index 27, whereas $[\mathcal{O} : 3\mathcal{O}] = 9$; in particular $\mathfrak{p}$ is not invertible ($\mathfrak{p}^2 = 3\mathfrak{p}$ with $\mathfrak{p} \neq 3\mathcal{O}$ forbids cancellation).

Remark ($p = 2$ and hypotheses in the Lean). The structural statements (Lemma 3.2, Proposition 3.3, the shape trichotomy, and Proposition 3.6) hold for all p , including $p = 2$; the Legendre-symbol form is stated for odd p and becomes p -uniform with the Kronecker symbol $\left(\frac{d}{2}\right)$. The current Lean declarations `prime_inert_iff`, `prime_ramified_iff`, `prime_split_iff` carry a $p \neq 2$ hypothesis throughout (Mathlib’s `legendreSym` requires odd p); the equivalence “ $p\mathcal{O}$ not radical $\iff p \mid d$ ” is in fact true for $p = 2$ as well (Step 4 of the trichotomy proof), a possible future strengthening. Nothing in this section uses $d < 0$.

Definition 3.7 (Square-root count modulo a prime power).

Lean: `QuadraticOrder.cardSqrts`

✓Lean-verified

Uses: §1.1

Let p be a rational prime and $s \geq 1$. For $c \in \mathbb{Z}/p^s$ set

$$N_s(c) := \#\{x \in \mathbb{Z}/p^s : x^2 = c\}.$$

Here v_p denotes the p -adic valuation on $\mathbb{Z}$ (with $v_p(0) = \infty$), and for odd p and $\gcd(u, p) = 1$ we write $\left(\frac{u}{p}\right)$ for the Legendre symbol.

Lemma 3.8 (Lemma 3.2.6, corrected and completed).

Lean: `QuadraticOrder.cardSqrts_prime`, `QuadraticOrder.cardSqrts_prime_pow_coprime`, `QuadraticOrder.cardSqrts_prime_pow_even_val`, `QuadraticOrder.cardSqrts_odd_val_eq_zero`, `QuadraticOrder.cardSqrts_zero`, `QuadraticOrder.cardSqrts_two`, `QuadraticOrder.cardSqrts_four_odd`, `QuadraticOrder.cardSqrts_eight`, `QuadraticOrder.cardSqrts_two_pow_even_val`

✓ *Lean-verified*

Uses: §3.7, §1.1

Let p be a prime, $s \geq 1$ and $r \geq 0$ integers, and $u \in \mathbb{Z}$ with $\gcd(u, p) = 1$.

(i) (Even valuation.) Assume $2r < s$. If p is odd, then

$$N_s(p^{2r}u) = \begin{cases} 2p^r & \text{if } \left(\frac{u}{p}\right) = 1, \\ 0 & \text{if } \left(\frac{u}{p}\right) = -1, \end{cases}$$

where the condition $\left(\frac{u}{p}\right) = 1$, i.e. u a quadratic residue modulo p , is equivalent to u being a quadratic residue modulo p^{s-2r} (and modulo every positive power of p). If $p = 2$, the count is governed by the difference $s - 2r$, not by s :

$$N_s(2^{2r}u) = \begin{cases} 2^r & \text{if } s - 2r = 1 \text{ (no condition on } u), \\ 2^{r+1} & \text{if } s - 2r = 2 \text{ and } u \equiv 1 \pmod{4}, \\ 2^{r+2} & \text{if } s - 2r \geq 3 \text{ and } u \equiv 1 \pmod{8}, \\ 0 & \text{if } s - 2r = 2, u \equiv 3 \pmod{4}, \text{ or } s - 2r \geq 3, u \not\equiv 1 \pmod{8}. \end{cases}$$

(ii) (Odd valuation.) If $2r + 1 < s$, then $N_s(p^{2r+1}u) = 0$, for every prime p including $p = 2$.

(iii) (Zero.) $N_s(0) = p^{\lfloor s/2 \rfloor}$.

Moreover, every $c \in \mathbb{Z}/p^s$ falls under exactly one of (i)–(iii): a nonzero c has a well-defined valuation $v := v_p(c) < s$ and unit part u , and the hypotheses $2r < s$, resp. $2r + 1 < s$, are automatic; $c = 0$ is (iii). All cases of (i) are instances of the reduction identity

$$N_s(p^{2r}u) = p^r \cdot N_{s-2r}(u) \quad (2r < s, \gcd(u, p) = 1). \quad (*)$$

Proof.

Uses: §3.7

Step 1: the reduction identity (*). Assume $2r < s$, $\gcd(u, p) = 1$, and put $t := s - 2r \geq 1$. Consider

$$\rho : \mathbb{Z}/p^{s-r} \twoheadrightarrow \mathbb{Z}/p^t \quad \text{and} \quad \mu : \mathbb{Z}/p^{s-r} \rightarrow \mathbb{Z}/p^s,$$

where ρ is reduction (well defined since $t \leq s - r$) and $\mu(b) := p^r \tilde{b} \bmod p^s$ for any integer lift $\tilde{b}$ of b (well defined: replacing $\tilde{b}$ by $\tilde{b} + p^{s-r}c$ changes $p^r \tilde{b}$ by $p^s c$).

μ is injective: $\mu(b) = \mu(b')$ gives $p^s \mid p^r(\tilde{b} - \tilde{b}')$, hence $p^{s-r} \mid \tilde{b} - \tilde{b}'$, i.e. $b = b'$.

Every solution of $x^2 = p^{2r}u$ lies in the image of μ : let a be an integer lift of a solution x , so $p^s \mid a^2 - p^{2r}u$. If $v_p(a) \leq r-1$, then $v_p(a^2) = 2v_p(a) \leq 2r-2 < 2r = v_p(p^{2r}u)$, so by the ultrametric equality for distinct valuations $v_p(a^2 - p^{2r}u) = 2v_p(a) < 2r < s$, a contradiction. Hence $v_p(a) \geq r$, and writing $a = p^r\tilde{b}$ we get $x = \mu(b)$ with $b := \tilde{b} \bmod p^{s-r}$.

Solution criterion: for $b \in \mathbb{Z}/p^{s-r}$ with lift $\tilde{b}$,

$$\mu(b)^2 = p^{2r}u \text{ in } \mathbb{Z}/p^s \iff p^s \mid p^{2r}(\tilde{b}^2 - u) \iff p^t \mid \tilde{b}^2 - u \iff \rho(b)^2 = u \text{ in } \mathbb{Z}/p^t,$$

the last step because $\tilde{b}^2 \bmod p^t$ depends only on $\rho(b)$. Therefore μ restricts to a bijection from $\rho^{-1}(\{y \in \mathbb{Z}/p^t : y^2 = u\})$ onto $\{x \in \mathbb{Z}/p^s : x^2 = p^{2r}u\}$. Since ρ is a surjective homomorphism of cyclic groups of orders p^{s-r} and p^t , each fiber has exactly $p^{(s-r)-t} = p^r$ elements, proving (*).

Step 2: square roots of a unit modulo p^t , $t \geq 1$. Any solution of $y^2 = u$ with $\gcd(u, p) = 1$ is a unit (else $p \mid u$). So the solutions lie in the abelian group $G := (\mathbb{Z}/p^t)^\times$ of order $\varphi(p^t) = p^{t-1}(p-1)$. Let $\sigma : G \rightarrow G$, $\sigma(y) = y^2$. Fibers of the homomorphism σ over its image are cosets of $\ker \sigma$; hence

$$N_t(u) = \begin{cases} |\ker \sigma| & u \in \sigma(G), \\ 0 & u \notin \sigma(G), \end{cases} \quad \text{and} \quad |\sigma(G)| = |G|/|\ker \sigma|. \quad (2.1)$$

(a) *Kernel, p odd.* If $y^2 = 1$ with odd-prime modulus, an integer lift $\tilde{y}$ satisfies $p^t \mid (\tilde{y}-1)(\tilde{y}+1)$; since $\gcd(\tilde{y}-1, \tilde{y}+1) \mid 2$ and p is odd, p^t divides one factor, so $y = \pm 1$, distinct as $p^t \geq 3$. Thus $|\ker \sigma| = 2$.

(b) *Image, p odd.* Claim: $u \in \sigma(G) \iff \left(\frac{u}{p}\right) = 1$. The reduction $\pi : (\mathbb{Z}/p^t)^\times \rightarrow (\mathbb{Z}/p)^\times$ is surjective with $|\ker \pi| = p^{t-1}$ odd; in a group of odd order q every element is a square, $h = (h^{(q+1)/2})^2$. If $u = v^2$ then $\pi(u) = \pi(v)^2$ is a square mod p . Conversely if $\pi(u) = \bar{w}^2$, pick w with $\pi(w) = \bar{w}$; then $uw^{-2} \in \ker \pi$ is a square v^2 , so $u = (wv)^2$. With (a) and (2.1): $N_t(u) = 2$ if $\left(\frac{u}{p}\right) = 1$, else 0; in particular being a quadratic residue mod p^t is, for units, independent of $t \geq 1$. (Equivalently: the two simple roots mod p lift uniquely through each modulus by Hensel's lemma, the route taken by the Lean proof.)

(c) $p = 2$, $t = 1$. In $\mathbb{Z}/2$, odd u means $u = 1$ and $y^2 = y$, so $N_1(u) = 1$ unconditionally.

(d) $p = 2$, $t = 2$. $(\mathbb{Z}/4)^\times = \{1, 3\}$ with $1^2 = 3^2 = 1$: $N_2(u) = 2$ if $u \equiv 1 \pmod{4}$, else 0.

(e) *Kernel*, $p = 2$, $t \geq 3$. If $y^2 = 1$, an odd lift $\tilde{y}$ satisfies $2^t \mid (\tilde{y}-1)(\tilde{y}+1)$, a product of consecutive even integers of which exactly one is $\equiv 2 \pmod{4}$; the other therefore has 2-adic valuation $\geq t-1$, so $y \in \{\pm 1, 2^{t-1} \pm 1\}$. Conversely $(\varepsilon + c2^{t-1})^2 = 1 + \varepsilon c 2^t + c^2 2^{2t-2} \equiv 1 \pmod{2^t}$ for $\varepsilon = \pm 1$, $c \in \{0, 1\}$, since $2t-2 \geq t$. The four elements are pairwise distinct mod 2^t for $t \geq 3$ (pairwise differences 2, 2^{t-1} , $2^{t-1} \pm 2$). Thus $|\ker \sigma| = 4$.

(f) *Image*, $p = 2$, $t \geq 3$. Every odd $\tilde{y} = 2z+1$ has $\tilde{y}^2 = 4z(z+1)+1 \equiv 1 \pmod{8}$ since $z(z+1)$ is even, so $\sigma(G) \subseteq H := \{u \in G : u \equiv 1 \pmod{8}\}$. By (2.1) and (e), $|\sigma(G)| = 2^{t-1}/4 = 2^{t-3}$, while $|H| = 2^t/8 = 2^{t-3}$; hence $\sigma(G) = H$ and $N_t(u) = 4$ if $u \equiv 1 \pmod{8}$, else 0.

Step 3: part (i). Combine (*) with Step 2 at $t = s - 2r$: for odd p , $N_s(p^{2r}u) = 2p^r$ or 0 according to $\left(\frac{u}{p}\right)$; for $p = 2$ and $t = 1, 2, \geq 3$ respectively, $N_s(2^{2r}u) = 2^r \cdot 1$, $2^r \cdot 2[u \equiv 1 \pmod{4}]$, $2^r \cdot 4[u \equiv 1 \pmod{8}]$, as displayed. (For $t = 1$ the condition $u \equiv 1 \pmod{2}$ is automatic.)

Step 4: part (ii). Assume $2r+1 < s$ and let a lift a putative solution, so $p^s \mid a^2 - p^{2r+1}u$. Then $p^{2r+1} \mid a^2$, so $2v_p(a) \geq 2r+1$ and, being even, $2v_p(a) \geq 2r+2 > 2r+1 = v_p(p^{2r+1}u)$. The ultrametric equality for distinct valuations gives $v_p(a^2 - p^{2r+1}u) = 2r+1 < s$ (also when $a = 0$, with $v_p(a^2) = \infty$), contradicting $p^s \mid a^2 - p^{2r+1}u$. Hence no solutions exist.

Step 5: part (iii). Set $\kappa := \lceil s/2 \rceil$, so $s - \kappa = \lfloor s/2 \rfloor$. If $x = p^\kappa y$ then $x^2 = p^{2\kappa} y^2 = 0$ since $2\kappa \geq s$. Conversely $x^2 = 0$ with lift a gives $2v_p(a) \geq s$, hence $v_p(a) \geq \kappa$ as $v_p(a)$ is an integer (or ∞), so $x \in p^\kappa(\mathbb{Z}/p^s)$. Multiplication by p^κ on $\mathbb{Z}/p^s$ has kernel of order p^κ , so its image — the solution set — has order $p^s/p^\kappa = p^{\lfloor s/2 \rfloor}$. $\square$

Remark 3.9 (Divergences from the thesis).

Uses: §3.8

(1) *Glyph loss (ERRATA E6.5)*. The thesis display prints “ $2p^r$ ”, when $p = 2$ ”; the intended condition is $p \neq 2$. (2) *Statement form (ERRATA E6.5)*. The thesis routes the count through “ u a quadratic residue mod p^{s-2r} ” plus a count of $\tilde{x} \in \mathbb{Z}/p^{s-r}$ with $\tilde{x}^2 \equiv 1 \pmod{p^{s-2r}}$, i.e. through a chosen square root of u . For $p = 2$ the residue condition must be parsed as $u \equiv 1 \pmod{\min(2^{s-2r}, 8)}$; the statement above gives the counts directly with explicit congruence conditions. (3) *Completion*. Parts (ii) and (iii) are stated in the thesis only inline in the proof of Theorem 3.1.2 (§3.3): the zero case as the stratum count $p^{k-\lceil m/2 \rceil}$, the odd-valuation case in the ramified argument. They are promoted to explicit parts here, and the three parts jointly compute $N_s(c)$ for every $c \in \mathbb{Z}/p^s$.

Remark 3.10 (Critical warning: the condition is on $s - 2r$, not s (ERRATA E3)).

Uses: §3.8

For $p = 2$ the case split in part (i) is governed by $s - 2r$. In §3.3 of the thesis the lemma is applied with the modulus exponent $s = 2k - m$ varying over the ideal strata while r is fixed ($r = w - 1$ for D odd, $r = w$ for $D \equiv 4 \pmod{8}$, where $w = v_2(f)$). The thesis proof instead tested “ $2k - m \in \{1, 2\}$ ”, i.e. read the condition as one on s ; the two readings agree only when $r = 0$ (i.e. $w = 1$ in the D -odd cases). The dropped strata contribute nonzero counts (2^{w-1} or 2^w for D odd; 2^w for $D \equiv 4 \pmod{8}$, m odd) for *every* w , which is the root cause of the two false 2-adic cases of thesis Theorem 3.1.2 (ERRATA E1, E2), confirmed by explicit enumeration at $d = -48, -192, -16$. With the corrected counts, the $p = 2$ case of Theorem 3.1.2 coincides with the odd- p formulas evaluated at $p = 2$ (ERRATA E4). Any use of this lemma must bind r and s explicitly before case-splitting.

Lemma 3.11 (Localization at the unique prime over p).

Lean: `QuadraticOrder.exists_int_denominator, QuadraticOrder.isUnit_algebraMap_atPrime_iff_not_dvd_norm, QuadraticOrder.existsUnique_isPrime_mem_of_dvd_conductor`

Uses: §1.1, §3.3

Let $d = Df^2 < 0$ be an imaginary quadratic discriminant with D fundamental and $f \geq 1$ the conductor, and let $\mathcal{O} = \mathcal{O}_d = \mathbb{Z}[\tau]$, $\tau = (d + \sqrt{d})/2$, with $g(x) = x^2 - dx + \frac{d^2-d}{4}$ the minimal polynomial of τ , $N(\alpha) = \alpha\bar{\alpha}$ the norm, and $\mathbb{Z}_{(p)} = \{a/t : a, t \in \mathbb{Z}, p \nmid t\}$. Assume p is a rational prime with $p \mid f$. Then:

- (0) $\mathcal{O}$ has exactly one prime ideal $\mathfrak{P}$ lying over p , namely $\mathfrak{P} = (p, \tau - A_0)$ where $x - A_0$ is the unique monic irreducible factor of $g \pmod{p}$ ($A_0 = 0$ for odd p ; $A_0 \equiv d/4 \pmod{2}$ for $p = 2$); in particular p is not split in $\mathcal{O}$ and $\mathcal{O}/\mathfrak{P} \cong \mathbb{F}_p$.
- (i) (Integer denominators.) Every z in the localization $\mathcal{O}_{\mathfrak{P}} = \{\alpha/\beta : \alpha \in \mathcal{O}, \beta \in \mathcal{O} \setminus \mathfrak{P}\} \subseteq K = \mathbb{Q}(\sqrt{d})$ can be written $z = \alpha/t$ with $\alpha \in \mathcal{O}$, $t \in \mathbb{Z}$, $p \nmid t$; equivalently, inside K ,

$$\mathcal{O}_{\mathfrak{P}} = \mathbb{Z}_{(p)}[\tau] = \mathbb{Z}_{(p)} + \mathbb{Z}_{(p)}\tau.$$

- (ii) (Unit criterion.) For $\alpha = x + y\tau \in \mathcal{O}$, the image of α in $\mathcal{O}_{\mathfrak{P}}$ is a unit if and only if $p \nmid N(\alpha) = x^2 + dxy + \frac{d^2-d}{4}y^2$ in $\mathbb{Z}$.

Proof.

Uses: §3.3

Step 0 (uniqueness of the prime over p). By Proposition 3.3, the primes of $\mathcal{O}$ containing p are exactly the ideals $p\mathcal{O} + (g_i(\tau))$ where $\bar{g}_i$ runs over the monic irreducible factors of $\bar{g} := g \bmod p$ in $\mathbb{F}_p[x]$. We claim $\bar{g}$ is the square of a monic linear polynomial. If p is odd, $p \mid f$ gives $p^2 \mid d$, and $v_p(\frac{d^2-d}{4}) = v_p(d(d-1)) = v_p(d) \geq 2$ since $v_p(4) = 0$ and $p \nmid d-1$; hence $\bar{g} = x^2$. If $p = 2$, then $2 \mid f$ gives $4 \mid f^2 \mid d$, the linear term $-dx$ vanishes mod 2, and the constant term satisfies $\frac{d^2-d}{4} = \frac{d}{4}(d-1) \equiv \frac{d}{4} \pmod{2}$ because $d-1$ is odd; hence $\bar{g} = x^2$ when $8 \mid d$ and $\bar{g} = x^2 + 1 = (x+1)^2$ when $v_2(d) = 2$. In all cases $\bar{g} = (x - \bar{A}_0)^2$ has a single monic irreducible factor, so $\mathfrak{P} = (p, \tau - A_0)$ is the unique prime of $\mathcal{O}$ containing p , with $\mathcal{O}/\mathfrak{P} \cong \mathbb{F}_p[x]/(x - \bar{A}_0) \cong \mathbb{F}_p$. Any prime containing p lies over p : $\mathfrak{P} \cap \mathbb{Z}$ is a proper ideal of $\mathbb{Z}$ containing the maximal ideal $p\mathbb{Z}$, hence equals $p\mathbb{Z}$. This proves (0); in particular p is not split in $\mathcal{O}$.

Step 1 ($\mathfrak{P}$ is stable under conjugation). Conjugation $\iota : \alpha \mapsto \bar{\alpha}$ restricts to a ring automorphism of $\mathcal{O}$, since $\iota(x + y\tau) = (x + yd) - y\tau \in \mathcal{O}$ and ι is an involutive field automorphism of K . Thus $\iota(\mathfrak{P})$ is a prime ideal of $\mathcal{O}$ containing $\iota(p) = p$, and the uniqueness of Step 0 gives $\iota(\mathfrak{P}) = \mathfrak{P}$.

Step 2 (norm detects membership in $\mathfrak{P}$). We claim: for $\alpha \in \mathcal{O}$, $\alpha \in \mathfrak{P} \iff p \mid N(\alpha)$. ($\Rightarrow$) If $\alpha \in \mathfrak{P}$ then $\bar{\alpha} \in \iota(\mathfrak{P}) = \mathfrak{P}$ by Step 1, so $N(\alpha) = \alpha\bar{\alpha} \in \mathfrak{P} \cap \mathbb{Z} = p\mathbb{Z}$. ($\Leftarrow$) If $p \mid N(\alpha)$ then $\alpha\bar{\alpha} \in p\mathbb{Z} \subseteq \mathfrak{P}$; since $\mathfrak{P}$ is prime, $\alpha \in \mathfrak{P}$ or $\bar{\alpha} \in \mathfrak{P}$, and in the latter case applying ι gives $\alpha = \iota(\bar{\alpha}) \in \iota(\mathfrak{P}) = \mathfrak{P}$.

Step 3 (proof of (ii)). Since $\mathcal{O}_{\mathfrak{P}} \subseteq K$ with K a field, an element of $\mathcal{O}_{\mathfrak{P}}$ is a unit iff it is nonzero and its inverse in K lies in $\mathcal{O}_{\mathfrak{P}}$. If $\alpha \notin \mathfrak{P}$, then $\alpha \neq 0$ and $1/\alpha \in \mathcal{O}_{\mathfrak{P}}$ by definition, so α is a unit. If $\alpha \in \mathfrak{P}$ were a unit, write $1/\alpha = \gamma/\delta$ with $\gamma \in \mathcal{O}$, $\delta \notin \mathfrak{P}$; then $\delta = \alpha\gamma \in \mathfrak{P}$, a contradiction. Hence α is a unit in $\mathcal{O}_{\mathfrak{P}}$ iff $\alpha \notin \mathfrak{P}$, which by Step 2 holds iff $p \nmid N(\alpha)$. Expanding $N(x + y\tau)$ via $\tau + \bar{\tau} = d$ and $\tau\bar{\tau} = \frac{d^2-d}{4}$ gives the stated form.

Step 4 (proof of (i)). If $t \in \mathbb{Z}$ with $p \nmid t$, then $t \notin \mathfrak{P}$ (as $\mathfrak{P} \cap \mathbb{Z} = p\mathbb{Z}$), so $\alpha/t \in \mathcal{O}_{\mathfrak{P}}$ for all $\alpha \in \mathcal{O}$; this inclusion holds for any prime over any p . Conversely, let $z = \alpha/\beta \in \mathcal{O}_{\mathfrak{P}}$ with $\alpha \in \mathcal{O}$ and $\beta \in \mathcal{O} \setminus \mathfrak{P}$. By Step 2, $p \nmid N(\beta)$, and $N(\beta) \neq 0$ since $\beta \neq 0$. Then

$$z = \frac{\alpha}{\beta} = \frac{\alpha\bar{\beta}}{\beta\bar{\beta}} = \frac{\alpha\bar{\beta}}{N(\beta)},$$

with numerator $\alpha\bar{\beta} \in \mathcal{O}$ and integer denominator $t = N(\beta)$ prime to p . Finally, bringing coordinates to a common denominator, $\{\alpha/t : \alpha \in \mathcal{O}, t \in \mathbb{Z}, p \nmid t\} = \mathbb{Z}_{(p)} + \mathbb{Z}_{(p)}\tau$, and $\mathbb{Z}_{(p)}[\tau] = \mathbb{Z}_{(p)} + \mathbb{Z}_{(p)}\tau$ because τ satisfies the monic quadratic g over $\mathbb{Z} \subseteq \mathbb{Z}_{(p)}$. $\square$

Remark (failure for split p ; ERRATA E6.2). Both (i) and (ii) are *false* without the hypothesis $p \mid f$ (more precisely, without the uniqueness of the prime over p that it forces). For $d = -4$, $\mathcal{O} = \mathbb{Z}[i]$, $p = 5 = (2+i)(2-i)$, $\mathfrak{P} = (2+i)$: the element $2-i \notin \mathfrak{P}$ is a unit of $\mathcal{O}_{\mathfrak{P}}$ although $5 \mid N(2-i) = 5$, violating (ii); and $1/(2-i) = (2+i)/5 \in \mathcal{O}_{\mathfrak{P}}$ admits no integer denominator prime to 5 (from $t(2+i) = 5\alpha$ one gets $t = (2-i)\alpha$, whence $t^2 = 5N(\alpha)$ and $5 \mid t$), violating (i). The thesis states the lemma for an arbitrary prime with the hypothesis “ p not split in $\mathcal{O}$ ”; following ERRATA E6.2 the blueprint version carries the standing hypothesis $p \mid f$ under which non-splitness is proved (Step 0) rather than assumed. The thesis’s original proof of Step 2 factors $N(\beta) = [\mathcal{O} : \beta\mathcal{O}]$ into local indices over the primary components of $\beta\mathcal{O}$; the conjugation argument above is an equivalent, formalization-friendly replacement resting on the same pivot (uniqueness of the prime over p).

Throughout this node: $d < 0$ is an integer with $d \equiv 0, 1 \pmod{4}$, written $d = Df^2$ with D a fundamental discriminant and $f \geq 1$ the conductor; $\mathcal{O} = \mathcal{O}_d = \mathbb{Z}[\tau]$ with $\tau = (d + \sqrt{d})/2$, so that $\mathcal{O} \cong \mathbb{Z}[x]/(g)$ for $g(x) = x^2 - dx + \frac{d^2-d}{4} \in \mathbb{Z}[x]$ (integral since $4 \mid d^2 - d$), a free $\mathbb{Z}$ -module with basis $(1, \tau)$ and $\tau^2 = d\tau - q$, $q := \frac{d^2-d}{4}$. For $A \in \mathbb{Z}$ one has $N(\tau - A) = g(A)$. The index of a finite-index ideal is $[\mathcal{O} : \mathfrak{a}] := \#(\mathcal{O}/\mathfrak{a})$. We fix a rational prime $p \mid f$ and $m \geq 0$; $\mathfrak{p}$ denotes the unique prime of $\mathcal{O}$ above p (unique because $g \bmod p$ is the square of a linear polynomial when $p \mid f$; see the prime-classification node).

Definition 3.12 (Canonical ideal; admissible pair).

Lean: `QuadraticOrder.canonicalIdeal`, `QuadraticOrder.`

`CanonicalAdmissible`, `QuadraticOrder.canonicalZSpan`

Uses: §1.1

For integers $k \geq 0$ and A , the *canonical ideal* attached to (p, m, k, A) is

$$\mathfrak{a}_{k,A} := (p^k, p^{m-k}(\tau - A)) \subseteq \mathcal{O}.$$

The pair (k, A) is *admissible* (for p, m) if

$$\lceil m/2 \rceil \leq k \leq m \quad \text{and} \quad p^{2k-m} \mid g(A).$$

For integer k the bound $\lceil m/2 \rceil \leq k$ is equivalent to $m \leq 2k$; together with $k \leq m$ this makes both exponents $2k - m$ and $m - k$ nonnegative. Whether $p^{2k-m} \mid g(A)$ depends only on the class $[A] \in \mathbb{Z}/p^{2k-m}\mathbb{Z}$, since $g(A + tp^{2k-m}) \equiv g(A) \pmod{p^{2k-m}}$. We also write $L_{k,A} := \mathbb{Z}p^k + \mathbb{Z}p^{m-k}(\tau - A)$ for the $\mathbb{Z}$ -span of the two generators.

Lemma 3.13 ($\mathbb{Z}$ -spans suffice (Remark 3.2.5)).

Lean: `QuadraticOrder.canonicalIdeal_eq_zSpan`

Uses: §3.12

Let (k, A) be admissible. Then $\mathfrak{a}_{k,A} = L_{k,A}$: every element of the ideal is a $\mathbb{Z}$ -linear combination of the two generators.

Proof.

Uses: §3.12, §1.1

Clearly $L_{k,A} \subseteq \mathfrak{a}_{k,A}$; conversely it suffices to show $L_{k,A}$ is an ideal, and since $\mathcal{O} = \mathbb{Z} + \mathbb{Z}\tau$ and $L_{k,A}$ is an additive subgroup, it is enough that $\tau L_{k,A} \subseteq L_{k,A}$, checked on the two generators. Write $g(A) = Cp^{2k-m}$, $C \in \mathbb{Z}$ (admissibility). First,

$$\tau \cdot p^k = Ap^k + p^{2k-m} \cdot [p^{m-k}(\tau - A)],$$

since the right side is $Ap^k + p^k(\tau - A) = p^k\tau$; both coefficients are integers because $2k - m \geq 0$. Second, using $\tau^2 = d\tau - q$ and $(d - A)\tau - q = (d - A)(\tau - A) - g(A)$ (as $g(A) = q - (d - A)A$),

$$\tau \cdot p^{m-k}(\tau - A) = p^{m-k}((d - A)\tau - q) = (d - A) \cdot [p^{m-k}(\tau - A)] - Cp^k,$$

where $p^{m-k}g(A) = Cp^{m-k}p^{2k-m} = Cp^k$. Both are $\mathbb{Z}$ -combinations of the generators. (These are the two identities proved in the existing Lean scaffolding; the thesis's Remark 3.2.5 carries a sign slip in this computation, ERRATA E6.9, fixed here.) $\square$

Lemma 3.14 (Index of the canonical ideal).

Lean: `QuadraticOrder.idealIndex_canonicalIdeal`

Uses: §3.12, §3.13, §2.1

Let (k, A) be admissible. Then $[\mathcal{O} : \mathfrak{a}_{k,A}] = p^m$.

Proof.

Uses: §3.13, §2.1

By Lemma 3.13, $\mathfrak{a}_{k,A}$ is the subgroup of $\mathcal{O} \cong \mathbb{Z}^2$ spanned by the vectors with coordinates $(p^k, 0)$ and $(-Ap^{m-k}, p^{m-k})$ in the basis $(1, \tau)$, i.e. by the columns of $M = \begin{pmatrix} p^k & -Ap^{m-k} \\ 0 & p^{m-k} \end{pmatrix}$ with $\det M = p^m \neq 0$. Since M is triangular the index is computed directly: $(x, y) = a(p^k, 0) + b(-Ap^{m-k}, p^{m-k})$ for some $a, b \in \mathbb{Z}$ iff $p^{m-k} \mid y$ and $p^k \mid x + Ay$ (necessarily $b = y/p^{m-k}$, and the first coordinate then reads $x + Ay = ap^k$). Hence $\mathfrak{a}_{k,A}$ is the kernel of the surjective group homomorphism $\mathbb{Z}^2 \rightarrow \mathbb{Z}/p^k\mathbb{Z} \times \mathbb{Z}/p^{m-k}\mathbb{Z}$,

$(x, y) \mapsto (x + Ay, y)$ — surjective because $(x, y) \mapsto (x + Ay, y)$ is a bijection of $\mathbb{Z}^2$ — and $[\mathcal{O} : \mathfrak{a}_{k,A}] = p^k \cdot p^{m-k} = p^m = |\det M|$. (The index infrastructure supplies the definition of the index as an additive-subgroup index; the determinant formulation is how the Lean proof phrases the computation.) $\square$

Lemma 3.15 (Minimal p -power; bridge lemma E6.4b).

Lean: `QuadraticOrder.canonicalIdeal_minimal_pow`

Uses: §3.12, §3.13

Let (k, A) be admissible. Then $\mathfrak{a}_{k,A} \cap \mathbb{Z} = p^k \mathbb{Z}$. In particular $p^k \in \mathfrak{a}_{k,A}$ and, if $k \geq 1$, $p^{k-1} \notin \mathfrak{a}_{k,A}$; that is, $k = \min\{j \geq 0 : p^j \in \mathfrak{a}_{k,A}\}$.

Proof.

Uses: §3.13

By Lemma 3.13 a general element of $\mathfrak{a}_{k,A}$ is $ap^k + bp^{m-k}(\tau - A) = (ap^k - Abp^{m-k}) + (bp^{m-k})\tau$ with $a, b \in \mathbb{Z}$. By uniqueness of coordinates in the basis $(1, \tau)$ it lies in $\mathbb{Z}$ iff $bp^{m-k} = 0$, i.e. $b = 0$; hence $\mathfrak{a}_{k,A} \cap \mathbb{Z} = p^k \mathbb{Z}$, and $p^j \in \mathfrak{a}_{k,A}$ forces $k \leq j$.

This is the clause missing from the thesis's converse (ERRATA E6.4b): without it, ideals could be double-counted across different values of k in the decomposition $r(p^m) = \sum_k r_k(p^m)$ of §3.3. $\square$

Proposition 3.16 (Existence of the normal form).

Lean: `QuadraticOrder.exists_canonicalIdeal_eq`

Uses: §3.12, §2.1

Every ideal $\mathfrak{a} \subseteq \mathcal{O}$ with $[\mathcal{O} : \mathfrak{a}] = p^m$ equals $\mathfrak{a}_{k,A}$ for some admissible pair (k, A) , where necessarily $k = \min\{j \geq 0 : p^j \in \mathfrak{a}\}$.

Proof.

Uses: §3.12, §2.1

If $m = 0$ then $\mathfrak{a} = \mathcal{O} = (1, \tau) = \mathfrak{a}_{0,0}$, the pair $(0, 0)$ is admissible, and $\min\{j \geq 0 : p^j \in \mathcal{O}\} = 0 = k$. Assume $m \geq 1$.

Step 1 ($p^m \in \mathfrak{a}$). The additive group $\mathcal{O}/\mathfrak{a}$ has order p^m , so $p^m \cdot (1 + \mathfrak{a}) = 0$, i.e. $p^m \in \mathfrak{a}$ (Lagrange; packaged in the index infrastructure).

Step 2 (definition of k ; $1 \leq k \leq m$). $\mathfrak{a} \cap \mathbb{Z}$ is an ideal of $\mathbb{Z}$ containing $p^m \mathbb{Z}$, hence equals $p^k \mathbb{Z}$ for some $0 \leq k \leq m$, and $k = \min\{j : p^j \in \mathfrak{a}\}$. If $k = 0$ then $\mathfrak{a} = \mathcal{O}$, contradicting $[\mathcal{O} : \mathfrak{a}] > 1$; so $k \geq 1$.

Step 3 (reduction mod p^k). Let $R_k := (\mathbb{Z}/p^k \mathbb{Z})[x]/(\bar{g})$. Since $\bar{g}$ is monic of degree 2, Euclidean division shows every class of R_k has unique coordinates $c_1 \bar{x} + c_0$ with $c_i \in \mathbb{Z}/p^k \mathbb{Z}$; thus $\#R_k = p^{2k}$ and the constants $\mathbb{Z}/p^k \mathbb{Z} \hookrightarrow R_k$ embed. The map $\pi : \mathcal{O} \cong \mathbb{Z}[x]/(g) \twoheadrightarrow R_k$, $\tau \mapsto \bar{x}$, has kernel $p^k \mathcal{O} \subseteq \mathfrak{a}$. Set

$\mathfrak{b} := \pi(\mathfrak{a})$; then $\mathfrak{a} = \pi^{-1}(\mathfrak{b})$, $R_k/\mathfrak{b} \cong \mathcal{O}/\mathfrak{a}$, hence $\#\mathfrak{b} = p^{2k-m}$; in particular $m \leq 2k$, i.e. $\lceil m/2 \rceil \leq k$.

Step 4 (no nonzero constants). For $c \in \mathbb{Z}$: $\pi(c) \in \mathfrak{b}$ iff $c \in \mathfrak{a} + p^k\mathcal{O} = \mathfrak{a}$ iff $c \in p^k\mathbb{Z}$ iff $\pi(c) = 0$. So $\mathfrak{b} \cap (\mathbb{Z}/p^k\mathbb{Z}) \cdot 1 = 0$.

Step 5 (principal normal form in R_k). We claim: any ideal $\mathfrak{b} \subseteq R_k$ with no nonzero constants equals $(\mathbb{Z}/p^k\mathbb{Z}) \cdot \xi$ with $\xi = p^a(\bar{x} - \bar{A})$ for some $0 \leq a \leq k$ and $A \in \mathbb{Z}$ satisfying $g(A) \equiv 0 \pmod{p^{k-a}}$, and then $\#\mathfrak{b} = p^{k-a}$. If $\mathfrak{b} = 0$ take $a = k$, $A = 0$ ($\xi = 0$). Otherwise: some element has nonzero $\bar{x}$ -coordinate (else all elements are constants, so $\mathfrak{b} = 0$). Each nonzero $c \in \mathbb{Z}/p^k\mathbb{Z}$ can be written $u p^{v(c)}$ with u a unit and the exponent $v(c) \in \{0, \dots, k-1\}$ uniquely determined. Pick $\eta_0 \in \mathfrak{b}$ with nonzero $\bar{x}$ -coordinate minimizing $a := v(c_1(\eta_0))$, and normalize by a unit so that $\xi_0 = p^a \bar{x} + c'_0 \in \mathfrak{b}$. Then $p^{k-a}\xi_0 = p^{k-a}c'_0 \cdot 1 \in \mathfrak{b}$ is a constant, hence zero, so $c'_0 = -p^a \bar{A}$ for some $A \in \mathbb{Z}$, and $\xi := p^a(\bar{x} - \bar{A}) \in \mathfrak{b}$. Next, from $\bar{x}^2 = \bar{d}\bar{x} - \bar{q}$,

$$\bar{x}\xi = (\bar{d} - \bar{A})\xi - p^a \bar{g}(\bar{A}) \cdot 1,$$

so $p^a \bar{g}(\bar{A}) \cdot 1 \in \mathfrak{b}$ is a constant, hence zero: $g(A) \equiv 0 \pmod{p^{k-a}}$. For $\mathfrak{b} \subseteq (\mathbb{Z}/p^k\mathbb{Z})\xi$: given $\eta \in \mathfrak{b}$, if $c_1(\eta) = 0$ then $\eta = 0$; else $b := v(c_1(\eta)) \geq a$ by minimality, and subtracting the appropriate unit multiple $u'p^{b-a}\xi$ kills the $\bar{x}$ -coordinate, leaving a constant of $\mathfrak{b}$, which is zero — so $\eta = u'p^{b-a}\xi$. Finally $\#\mathfrak{b} = p^k/\#\text{Ann}(\xi) = p^{k-a}$, since $c\xi = 0$ iff $cp^a = 0$ iff $p^{k-a} \mid c$.

Step 6 (match and pull back). Comparing cardinalities, $p^{k-a} = p^{2k-m}$ gives $a = m - k$, and the root condition becomes $g(A) \equiv 0 \pmod{p^{2k-m}}$: the pair (k, A) is admissible. Since $\mathfrak{a} \supseteq \ker \pi$,

$$\mathfrak{a} = \pi^{-1}(\mathfrak{b}) = p^k\mathcal{O} + p^{m-k}(\tau - A)\mathcal{O} = \mathfrak{a}_{k,A}. \quad \square$$

Proposition 3.17 (Uniqueness of the parameters).

Lean: *QuadraticOrder.canonicalIdeal_inj*

Uses: §3.12, §3.13, §3.15

Let (k, A) and (k', A') be admissible (same p, m). Then

$$\mathfrak{a}_{k,A} = \mathfrak{a}_{k',A'} \iff k = k' \text{ and } A \equiv A' \pmod{p^{2k-m}}.$$

Proof.

Uses: §3.13, §3.15

($\Leftarrow$) If $A' = A + tp^{2k-m}$ then $p^{m-k}(\tau - A') = p^{m-k}(\tau - A) - tp^k$, so each generator of either ideal lies in the other and the ideals agree.

($\Rightarrow$) Suppose $\mathfrak{a} := \mathfrak{a}_{k,A} = \mathfrak{a}_{k',A'}$. By Lemma 3.15 applied to both presentations, $k = \min\{j : p^j \in \mathfrak{a}\} = k'$. By Lemma 3.13, $p^{m-k}(\tau - A') \in$

$\mathbb{Z}p^k + \mathbb{Z}p^{m-k}(\tau - A)$; write $p^{m-k}(\tau - A') = a_1p^k + a_2p^{m-k}(\tau - A)$ with $a_1, a_2 \in \mathbb{Z}$. Comparing τ -coordinates gives $a_2 = 1$; comparing constant coordinates gives $p^{m-k}(A - A') = a_1p^k$, hence $A - A' = a_1p^{2k-m}$. $\square$

Theorem 3.18 (Lemma 3.2.4: normal-form bijection and counting formula).

Lean: `QuadraticOrder.idealCount_prime_pow_eq_sum_rootCount`

Uses: §3.12, §3.14, §3.15, §3.16, §3.17, §3.3, §2.1

Fix $p \mid f$ and $m \geq 0$. The assignment $(k, [A]) \mapsto \mathfrak{a}_{k,A}$ is a bijection

$$\{(k, [A]) : \lceil m/2 \rceil \leq k \leq m, [A] \in \mathbb{Z}/p^{2k-m}\mathbb{Z}, g(A) \equiv 0 \pmod{p^{2k-m}}\} \xrightarrow{\sim} \{\mathfrak{a} \subseteq \mathcal{O} : [\mathcal{O} : \mathfrak{a}] = p^m\},$$

under which k is the minimal exponent with $p^k \in \mathfrak{a}_{k,A}$. Consequently

$$\#\{\mathfrak{a} \subseteq \mathcal{O} : [\mathcal{O} : \mathfrak{a}] = p^m\} = \sum_{k=\lceil m/2 \rceil}^m \#\{A \in \mathbb{Z}/p^{2k-m}\mathbb{Z} : g(A) \equiv 0 \pmod{p^{2k-m}}\}.$$

Moreover, for $m \geq 1$ the target set is exactly the set of $\mathfrak{p}$ -primary ideals of norm p^m , where $\mathfrak{p}$ is the unique prime of $\mathcal{O}$ above p .

Proof.

Uses: §3.14, §3.15, §3.16, §3.17, §3.3, §2.1

The map is well defined on classes and lands in the target by Proposition 3.17($\Leftarrow$) and Lemma 3.14; it is injective by Proposition 3.17($\Rightarrow$) — whose reduction to a common k is Lemma 3.15, the bridge clause E6.4b — and surjective by Proposition 3.16. The source is a finite disjoint union over k of root sets in finite rings, whence the displayed cardinality identity.

For the last sentence: an ideal of index p^m ($m \geq 1$) is $\mathfrak{p}$ -primary by bridge lemma (a) of the index infrastructure, whose proof rests on the uniqueness of the prime above $p \mid f$ (prime-classification node); conversely a $\mathfrak{p}$ -primary ideal of norm p^m has index p^m by definition. (For $m = 0$ the target is $\{\mathcal{O}\}$, matched by the single parameter $(0, [0])$; $\mathcal{O}$ itself is not a primary ideal, which is why the primary-ideal phrasing is restricted to $m \geq 1$ — a hygiene correction to the thesis statement.)

Refinement: with the §3.3 notation $r_k(p^m) = \#\{\mathfrak{a} : [\mathcal{O} : \mathfrak{a}] = p^m, p^k \in \mathfrak{a}, p^{k-1} \notin \mathfrak{a}\}$, the bijection restricts on each stratum k (by Lemma 3.15 and Proposition 3.16) to $r_k(p^m) = \#\{A \in \mathbb{Z}/p^{2k-m}\mathbb{Z} : g(A) \equiv 0\}$, the per- k identity consumed by the proof of Theorem 3.1.2. $\square$

Lemma 3.19 (Minimal p -power in a principal local ideal).

Lean: `QuadraticOrder.comap_map_span_tau_sub_eq_canonicalIdeal`

Uses: §1.1, §3.11, §3.18

Let $d < 0$ be a discriminant ($d \equiv 0, 1 \pmod{4}$), $d = Df^2$ with D fundamental and $f \geq 1$ the conductor, and let $\mathcal{O} = \mathbb{Z}[\tau]$ with $\tau = (d + \sqrt{d})/2$ be the order of discriminant d in $K = \mathbb{Q}(\sqrt{d})$, so that the minimal polynomial of τ is $g(x) = x^2 - dx + \frac{d^2-d}{4}$, the conjugate is $\bar{\tau} = d - \tau$, and $N(\tau - a) = g(a) > 0$ for all $a \in \mathbb{Z}$. Let p be a rational prime with $p \mid f$; then $g \bmod p$ is the square of a linear polynomial, so $\mathcal{O}$ has a unique prime ideal $\mathfrak{p}$ containing p , and the localization $\mathcal{O}_{\mathfrak{p}} \subseteq K$ satisfies $\mathcal{O}_{\mathfrak{p}} = \mathbb{Z}_{(p)}[\tau]$ by Lemma 3.11.

Let $r \geq 0$ be an integer, let $a \in \mathbb{Z}$, and set

$$\nu := v_p(N(\tau - a)) = v_p(g(a)) \geq 0.$$

Then the contraction to $\mathcal{O}$ of the principal ideal $p^r(\tau - a)\mathcal{O}_{\mathfrak{p}}$ is the canonical ideal of Lemma 3.18 with parameters $k = r + \nu$, $m = 2r + \nu$, $A = a$:

$$p^r(\tau - a)\mathcal{O}_{\mathfrak{p}} \cap \mathcal{O} = (p^{r+\nu}, p^r(\tau - a)).$$

In particular, every nonzero rational number lying in $p^r(\tau - a)\mathcal{O}_{\mathfrak{p}}$ has p -adic valuation at least $r + \nu$, and the smallest positive rational integer (equivalently, the smallest power of p) contained in $p^r(\tau - a)\mathcal{O}_{\mathfrak{p}}$ is exactly $p^{r+\nu}$.

Proof.

Uses: §1.1, §3.11, §3.18

Since $d < 0$, $g(a) = (a - d/2)^2 - d/4$ is a positive rational integer, so we may write $g(a) = p^\nu u$ with $u \in \mathbb{Z}_{>0}$ and $p \nmid u$; note $u \in \mathbb{Z}_{(p)}^\times$.

Step 1: quadratic relation. The elements $\tau - a$ and $\bar{\tau} - a$ have sum $d - 2a$ and product $N(\tau - a) = g(a)$, so

$$(\tau - a)^2 = (d - 2a)(\tau - a) - g(a). \quad (1)$$

Step 2: the local ring as a free module. By Lemma 3.11 and $p \mid f$ (which makes p non-split), $\mathcal{O}_{\mathfrak{p}} = \mathbb{Z}_{(p)}[\tau]$. Since $\tau^2 = d\tau - \frac{d^2-d}{4}$, we have $\mathbb{Z}_{(p)}[\tau] = \mathbb{Z}_{(p)} \oplus \mathbb{Z}_{(p)}\tau$, the sum being direct because $1, \tau$ are $\mathbb{Q}$ -linearly independent ($\tau \notin \mathbb{Q}$ as $d < 0$). Likewise $\mathcal{O} = \mathbb{Z} \oplus \mathbb{Z}\tau$, and coordinates with respect to the $\mathbb{Q}$ -basis $(1, \tau)$ of K are unique.

Step 3: normal form of the local ideal. We claim

$$p^r(\tau - a)\mathcal{O}_{\mathfrak{p}} = \mathbb{Z}_{(p)}p^{r+\nu} + \mathbb{Z}_{(p)}p^r(\tau - a). \quad (2)$$

For $\beta = e_0 + e_1\tau \in \mathcal{O}_{\mathfrak{p}}$ ($e_0, e_1 \in \mathbb{Z}_{(p)}$), write $\beta = (e_0 + ae_1) + e_1(\tau - a)$; then by (1),

$$p^r(\tau - a)\beta = \lambda \cdot p^r(\tau - a) + \mu \cdot p^{r+\nu}, \quad \lambda := e_0 + (d - a)e_1, \quad \mu := -ue_1. \quad (3)$$

The $\mathbb{Z}_{(p)}$ -linear map $(e_0, e_1) \mapsto (\lambda, \mu)$ has determinant $-u \in \mathbb{Z}_{(p)}^\times$, hence is a bijection of $\mathbb{Z}_{(p)}^2$. As the principal ideal consists exactly of the elements $p^r(\tau - a)\beta$ with $\beta \in \mathcal{O}_{\mathfrak{p}}$, claim (2) follows.

Step 4: containment $\supseteq$. Clearly $p^r(\tau - a)$ lies in both $\mathcal{O}$ and the local ideal. For the integer generator, $\bar{\tau} - a = (d - a) - \tau \in \mathcal{O}$ and $p^r(\tau - a)(\bar{\tau} - a) = p^r g(a) = p^{r+\nu}u$, so

$$p^{r+\nu} = p^r(\tau - a) \cdot u^{-1}(\bar{\tau} - a) \in p^r(\tau - a) \mathcal{O}_{\mathfrak{p}},$$

using $u^{-1} \in \mathbb{Z}_{(p)} \subseteq \mathcal{O}_{\mathfrak{p}}$. The contraction is an ideal of $\mathcal{O}$ containing both generators, hence contains $(p^{r+\nu}, p^r(\tau - a))$.

Step 5: containment $\subseteq$. Let $\xi \in p^r(\tau - a) \mathcal{O}_{\mathfrak{p}} \cap \mathcal{O}$. By (2), $\xi = \mu p^{r+\nu} + \lambda p^r(\tau - a)$ with $\lambda, \mu \in \mathbb{Z}_{(p)}$, and in the basis $(1, \tau)$,

$$\xi = (\mu p^{r+\nu} - a\lambda p^r) \cdot 1 + (\lambda p^r) \cdot \tau.$$

Since $\xi \in \mathcal{O} = \mathbb{Z} \oplus \mathbb{Z}\tau$ and coordinates are unique, $\lambda p^r \in \mathbb{Z}$ and $\mu p^{r+\nu} - a\lambda p^r \in \mathbb{Z}$. From $\lambda \in \mathbb{Z}_{(p)}$ we get $v_p(\lambda) \geq 0$, and from $\lambda p^r \in \mathbb{Z}$ we get $v_q(\lambda) \geq 0$ for every prime $q \neq p$; hence $\lambda \in \mathbb{Z}$. Then $a\lambda p^r \in \mathbb{Z}$ forces $\mu p^{r+\nu} \in \mathbb{Z}$, and the same argument gives $\mu \in \mathbb{Z}$. Thus ξ is a $\mathbb{Z}$ -linear (in particular $\mathcal{O}$ -linear) combination of $p^{r+\nu}$ and $p^r(\tau - a)$, so $\xi \in (p^{r+\nu}, p^r(\tau - a))$.

Steps 4 and 5 prove the contraction identity. The parameters are admissible for Lemma 3.18: with $k = r + \nu$ and $m = 2r + \nu$ we have $m/2 \leq k \leq m$ and $g(a) \equiv 0 \pmod{p^{2k-m}}$ since $2k - m = \nu = v_p(g(a))$.

Step 6: minimality. Let $\rho \in p^r(\tau - a) \mathcal{O}_{\mathfrak{p}} \cap \mathbb{Q}$ be nonzero. By (2), $\rho = \mu p^{r+\nu} + \lambda p^r(\tau - a)$ with $\lambda, \mu \in \mathbb{Z}_{(p)}$; its τ -coordinate λp^r must vanish, so $\lambda = 0$, $\rho = \mu p^{r+\nu}$, and $v_p(\rho) = v_p(\mu) + r + \nu \geq r + \nu$. Conversely, every $n \in \mathbb{Z}$ with $v_p(n) \geq r + \nu$ lies in the local ideal, since $n = (n/p^{r+\nu}) \cdot p^{r+\nu}$ with $n/p^{r+\nu} \in \mathbb{Z}_{(p)}$ and $p^{r+\nu}$ is in the ideal by Step 4. Hence the rational integers in the local ideal are exactly those with $v_p \geq r + \nu$, and the smallest positive one is $p^{r+\nu}$. $\square$

Lemma 3.20 (Equality criterion for principal local ideals).

Lean: `QuadraticOrder.map_span_tau_sub_eq_iff`

Uses: §1.1, §3.11, §3.19, §3.18

Let $d < 0$ be a quadratic discriminant ($d \equiv 0, 1 \pmod{4}$), let $\mathcal{O} = \mathcal{O}_d = \mathbb{Z}[\tau]$ with $\tau = \frac{d+\sqrt{d}}{2}$, and let $g(x) = x^2 - dx + \frac{d^2-d}{4}$ be the minimal polynomial of τ . Writing $\bar{\tau} := d - \tau$ for the conjugate of τ and N for the norm of $K = \mathbb{Q}(\sqrt{d})$, we have $N(\tau - t) = (\tau - t)(\bar{\tau} - t) = g(t) \neq 0$ for every $t \in \mathbb{Z}$. Write $d = Df^2$ with D fundamental and $f = \text{cond}(d)$, and let p be

a rational prime with $p \mid f$ (standing hypothesis of §3.2 of the thesis, made explicit per ERRATA E6.2). Let $\mathfrak{p}$ be the unique prime of $\mathcal{O}$ above p , so that $\mathcal{O}_{\mathfrak{p}} = \mathbb{Z}_{(p)}[\tau]$ by Lemma 3.11.

Then for all $\alpha, \beta \in \mathbb{Z}$ and all integers $r \geq 0$,

$$p^r(\tau - \alpha) \mathcal{O}_{\mathfrak{p}} = p^r(\tau - \beta) \mathcal{O}_{\mathfrak{p}} \iff v_p(g(\alpha)) = v_p(g(\beta)) \text{ and } \alpha \equiv \beta \pmod{p^{v_p(g(\alpha))}}.$$

The thesis proof is a two-line delegation to Lemma 3.19 and the uniqueness clause of Lemma 3.18; the proof below spells out the contraction normal-form identification that the delegation hides (Step 2), which a formal proof needs explicitly. The hypothesis $p \mid f$ is used only through the identification $\mathcal{O}_{\mathfrak{p}} = \mathbb{Z}_{(p)}[\tau]$: every step below happens inside $\mathbb{Z}_{(p)}[\tau]$, so the criterion for the mapped ideals $p^r(\tau - \alpha) \mathbb{Z}_{(p)}[\tau]$ holds for every prime p ; the Lean declaration `map_span_tau_sub_eq_iff` nevertheless follows this statement — it is phrased at the localization of $\mathcal{O}$ at $\mathfrak{p}$ and carries the standing hypothesis $p \mid f$ — because the statement about the localization of $\mathcal{O}$ at a prime ideal is false for split p (e.g. $d = -4$, $p = 5$, $\mathfrak{p} = (5, \tau)$: $(\tau - 1)\mathcal{O}_{\mathfrak{p}} = (\tau - 2)\mathcal{O}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$ although $v_5(g(1)) = 1 \neq 0 = v_5(g(2))$).

Proof.

Uses: §1.1, §3.11, §3.19, §3.18

Put $\nu := v_p(g(\alpha))$ and $\mu := v_p(g(\beta))$, both finite since $g(\alpha), g(\beta) \neq 0$ (g has no rational roots as $d < 0$), and write $g(\alpha) = p^\nu u$ with $u \in \mathbb{Z}$, $p \nmid u$.

Step 0 (bases). $\{1, \tau\}$ is a $\mathbb{Z}$ -basis of $\mathcal{O}$ and a $\mathbb{Q}$ -basis of $K = \mathbb{Q}(\sqrt{d})$; for $t \in \mathbb{Z}$ the unimodular change of basis shows the same for $\{1, \tau - t\}$. Hence $\mathcal{O} = \mathbb{Z} \oplus \mathbb{Z}(\tau - t)$ and $\mathcal{O}_{\mathfrak{p}} = \mathbb{Z}_{(p)} \oplus \mathbb{Z}_{(p)}(\tau - t)$, coordinates in K are unique, and an element of $\mathcal{O}_{\mathfrak{p}}$ lies in $\mathcal{O}$ iff both coordinates are integers. Also $\mathbb{Z} \cap p^j \mathbb{Z}_{(p)} = p^j \mathbb{Z}$ for all $j \geq 0$.

Step 1 (p-power membership). Since $(\tau - \alpha)(\bar{\tau} - \alpha) = N(\tau - \alpha) = g(\alpha) = p^\nu u$ with $\bar{\tau} - \alpha = (d - \alpha) - \tau \in \mathcal{O}$ and $u^{-1} \in \mathbb{Z}_{(p)}$, we get

$$p^\nu = (\tau - \alpha) \cdot u^{-1}(\bar{\tau} - \alpha) \in (\tau - \alpha) \mathcal{O}_{\mathfrak{p}}.$$

Step 2 (contraction in normal form). Set $k_\alpha := r + \nu$, $m_\alpha := 2r + \nu$ and $\mathfrak{a}_\alpha := p^{k_\alpha} \mathbb{Z} \oplus p^r(\tau - \alpha) \mathbb{Z} \subseteq \mathcal{O}$. We claim

$$p^r(\tau - \alpha) \mathcal{O}_{\mathfrak{p}} \cap \mathcal{O} = \mathfrak{a}_\alpha,$$

and that $\mathfrak{a}_\alpha$ is an $\mathcal{O}$ -ideal of index p^{m_α} with $\mathfrak{a}_\alpha \cap \mathbb{Z} = p^{k_\alpha} \mathbb{Z}$; that is, $\mathfrak{a}_\alpha$ is exactly the normal-form ideal $(p^k, p^{m-k}(\tau - A))$ of Lemma 3.18 with $k = k_\alpha$, $m = m_\alpha$ (so $m/2 \leq k \leq m$, $m - k = r$, $2k - m = \nu$) and residue class $[A] = [\alpha] \in \mathbb{Z}/p^\nu$, the root condition $g(\alpha) \equiv 0 \pmod{p^{2k-m}}$ holding by

definition of ν , and p^k being the minimal positive integer of the ideal as in Lemma 3.19.

($\supseteq$) $p^r(\tau - \alpha)$ lies in both sides since $r \geq 0$ and $\alpha \in \mathbb{Z}$; and $p^{k_\alpha} = p^r p^\nu \in p^r(\tau - \alpha)\mathcal{O}_{\mathfrak{p}} \cap \mathbb{Z}$ by Step 1. The left-hand side is an additive group, so it contains $\mathfrak{a}_\alpha$.

($\subseteq$) Let $\xi = p^r(\tau - \alpha)\gamma \in \mathcal{O}$ with $\gamma = e_0 + e_1(\tau - \alpha)$, $e_0, e_1 \in \mathbb{Z}_{(p)}$ (Step 0). From $\tau^2 = d\tau - \frac{d^2-d}{4}$ one computes $(\tau - \alpha)^2 = (d - 2\alpha)(\tau - \alpha) - g(\alpha)$, whence

$$\xi = \underbrace{-e_1 u p^{k_\alpha}}_{b_0} \cdot 1 + \underbrace{p^r(e_0 + e_1(d - 2\alpha))}_{b_1} \cdot (\tau - \alpha), \quad b_0 \in p^{k_\alpha}\mathbb{Z}_{(p)}, \quad b_1 \in p^r\mathbb{Z}_{(p)}.$$

As $\xi \in \mathcal{O}$, Step 0 gives $b_0, b_1 \in \mathbb{Z}$, hence $b_0 \in p^{k_\alpha}\mathbb{Z}$ and $b_1 \in p^r\mathbb{Z}$, so $\xi = (b_0 p^{-k_\alpha}) p^{k_\alpha} + (b_1 p^{-r}) p^r(\tau - \alpha) \in \mathfrak{a}_\alpha$ with integer coefficients.

$\mathfrak{a}_\alpha$ is an ideal: it is τ -stable by

$$\tau p^{k_\alpha} = \alpha p^{k_\alpha} + p^\nu [p^r(\tau - \alpha)], \quad \tau p^r(\tau - \alpha) = -u p^{k_\alpha} + (d - \alpha) [p^r(\tau - \alpha)],$$

all coefficients integral (this is Remark 3.2.5 of the thesis for these generators). Its index is $p^{k_\alpha} \cdot p^r = p^{m_\alpha}$ (diagonal basis matrix in the basis $\{1, \tau - \alpha\}$). Its integers: an element $c_1 p^{k_\alpha} + c_2 p^r(\tau - \alpha)$ is rational iff $c_2 = 0$, so $\mathfrak{a}_\alpha \cap \mathbb{Z} = p^{k_\alpha}\mathbb{Z}$; in particular the minimal positive integer of $p^r(\tau - \alpha)\mathcal{O}_{\mathfrak{p}}$ is $p^{r+v_p(N(\tau-\alpha))}$, recovering Lemma 3.19. This proves the claim.

Step 3 (sufficiency). Assume $\nu = \mu$ and $\beta = \alpha + t p^\nu$, $t \in \mathbb{Z}$. By Step 1, $\tau - \beta = (\tau - \alpha) - t p^\nu \in (\tau - \alpha)\mathcal{O}_{\mathfrak{p}}$, so $(\tau - \beta)\mathcal{O}_{\mathfrak{p}} \subseteq (\tau - \alpha)\mathcal{O}_{\mathfrak{p}}$; exchanging α and β (legitimate since $v_p(g(\beta)) = \nu$ as well) gives the reverse inclusion, and multiplying by p^r concludes.

Step 4 (necessity). Assume $J := p^r(\tau - \alpha)\mathcal{O}_{\mathfrak{p}} = p^r(\tau - \beta)\mathcal{O}_{\mathfrak{p}}$.

(i) By Lemma 3.19 (or Step 2), the minimal positive integer of J equals both $p^{r+\nu}$ and $p^{r+\mu}$, so $\nu = \mu$. If $\nu = 0$ the congruence is modulo $p^0 = 1$ and there is nothing more to prove; assume $\nu \geq 1$ and set $k := r + \nu$.

(ii) Intersecting J with $\mathcal{O}$ and applying Step 2 on each side,

$$p^k\mathbb{Z} \oplus p^r(\tau - \alpha)\mathbb{Z} = J \cap \mathcal{O} = p^k\mathbb{Z} \oplus p^r(\tau - \beta)\mathbb{Z}.$$

In particular $p^r(\tau - \alpha) = a_1 p^k + a_2 p^r(\tau - \beta)$ for some $a_1, a_2 \in \mathbb{Z}$ — the “ $\mathcal{O}$ -combinations are $\mathbb{Z}$ -combinations” step that Remark 3.2.5 supplies inside Lemma 3.18, immediate here because the contraction is computed as a $\mathbb{Z}$ -span. Comparing τ -coordinates in the basis $\{1, \tau\}$ gives $p^r = a_2 p^r$, so $a_2 = 1$; the constant coordinates then give $p^r(\beta - \alpha) = a_1 p^k$, i.e. $\beta - \alpha = a_1 p^{k-r} = a_1 p^\nu$, which is the uniqueness computation of Lemma 3.18 applied to the two normal-form presentations of $J \cap \mathcal{O}$. Hence $\alpha \equiv \beta \pmod{p^\nu}$. $\square$

Proposition 3.21 (Ideal counts in the maximal order).

Lean: `QuadraticOrder.maximalIdealCount_split`, `QuadraticOrder.maximalIdealCount_inert`, `QuadraticOrder.maximalIdealCount_ramified`

Uses: §1.1

Let K be a quadratic field — real or imaginary — of discriminant D (a fundamental discriminant), with ring of integers $\mathcal{O}_K = \mathbb{Z}[\tau_K]$, $\tau_K = \frac{D+\sqrt{D}}{2}$. For a nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ write $N(\mathfrak{a}) = [\mathcal{O}_K : \mathfrak{a}]$ for the index of $\mathfrak{a}$ in the additive group of $\mathcal{O}_K$. Let p be a rational prime, let $m \geq 0$ be an integer, and let $\left(\frac{D}{p}\right)$ denote the Kronecker symbol at p : the Legendre symbol of D for odd p , and, for $p = 2$, equal to 0 if D is even, 1 if $D \equiv 1 \pmod{8}$, and -1 if $D \equiv 5 \pmod{8}$. Then

$$\#\{\mathfrak{a} \subseteq \mathcal{O}_K \text{ ideal} : N(\mathfrak{a}) = p^m\} = \sum_{j=0}^m \left(\frac{D}{p}\right)^j$$

(convention $0^0 = 1$); explicitly, the count equals

$$\begin{cases} m+1 & \text{if } \left(\frac{D}{p}\right) = 1 \text{ (} p \text{ split),} \\ 1 & \text{if } \left(\frac{D}{p}\right) = -1 \text{ and } m \text{ even (} p \text{ inert), or } \left(\frac{D}{p}\right) = 0 \text{ (} p \text{ ramified),} \\ 0 & \text{if } \left(\frac{D}{p}\right) = -1 \text{ and } m \text{ odd (} p \text{ inert).} \end{cases}$$

Here p splits, is inert, resp. ramifies in $\mathcal{O}_K$ according as $p\mathcal{O}_K = \mathfrak{p}_1\mathfrak{p}_2$ with $\mathfrak{p}_1 \neq \mathfrak{p}_2$ prime, $p\mathcal{O}_K$ is prime, resp. $p\mathcal{O}_K = \mathfrak{p}^2$; the proof shows the trichotomy is detected by $\left(\frac{D}{p}\right)$ as indicated. (The thesis states this for imaginary quadratic K only; by ERRATA E6.9 imaginarity is not needed.)

Proof.

Uses: §1.1, §3.3

Write $\mathcal{O} := \mathcal{O}_K = \mathbb{Z}[\tau] \cong \mathbb{Z}[x]/(g)$ with $\tau := \tau_K$ and $g(x) := x^2 - Dx + \frac{D^2-D}{4}$, the minimal polynomial of τ ; note $\text{disc}(g) = D$. Bars denote reduction mod p .

Dedekind toolkit. $\mathcal{O}_K$ is a Dedekind domain; we use the following standard facts. (i) Every nonzero ideal factors uniquely as a finite product of nonzero primes, all nonzero primes are maximal, and $\mathfrak{a} \subseteq \mathfrak{b}$ iff $\mathfrak{b} \mid \mathfrak{a}$. (ii) Every nonzero ideal $\mathfrak{a}$ has finite index: for $0 \neq \alpha \in \mathfrak{a}$ the nonzero integer $n = |\alpha\bar{\alpha}| = |N_{K/\mathbb{Q}}(\alpha)|$ lies in $\mathfrak{a}$, so $n\mathcal{O} \subseteq \mathfrak{a}$ and $[\mathcal{O} : \mathfrak{a}] \leq n^2$. Consequently an ideal of norm p^m is automatically nonzero. (iii) If $\mathfrak{a} + \mathfrak{b} = \mathcal{O}$ then $\mathfrak{a}\mathfrak{b} = \mathfrak{a} \cap \mathfrak{b}$ and $\mathcal{O}/\mathfrak{a}\mathfrak{b} \cong \mathcal{O}/\mathfrak{a} \times \mathcal{O}/\mathfrak{b}$ (CRT), so $N(\mathfrak{a}\mathfrak{b}) = N(\mathfrak{a})N(\mathfrak{b})$; and powers of

distinct nonzero primes are comaximal (a maximal ideal containing $\mathfrak{q}^a + \mathfrak{q}^b$ would divide both $\mathfrak{q}^a$ and $\mathfrak{q}^b$, forcing $\mathfrak{q} = \mathfrak{q}'$). (iv) $N(\mathfrak{q}^k) = N(\mathfrak{q})^k$ for a nonzero prime $\mathfrak{q}$: in the filtration $\mathcal{O} \supseteq \mathfrak{q} \supseteq \cdots \supseteq \mathfrak{q}^k$ each quotient $\mathfrak{q}^i/\mathfrak{q}^{i+1}$ is an $\mathcal{O}/\mathfrak{q}$ -vector space whose subspaces are the ideals between $\mathfrak{q}^{i+1}$ and $\mathfrak{q}^i$; by (i) these are only $\mathfrak{q}^i, \mathfrak{q}^{i+1}$ themselves, and $\mathfrak{q}^{i+1} \subsetneq \mathfrak{q}^i$, so each quotient is one-dimensional of cardinality $N(\mathfrak{q})$. (v) A nonzero prime $\mathfrak{q}$ satisfies $\mathfrak{q} \cap \mathbb{Z} = (q)$ for a rational prime q , and $N(\mathfrak{q}) \in \{q, q^2\}$, since $\mathcal{O}/\mathfrak{q}$ is a field quotient of $\mathcal{O}/q\mathcal{O}$, which has order q^2 .

Step 1: primes above p and the trichotomy. By Proposition 3.3 applied to $\mathcal{O} = \mathbb{Z}[x]/(g)$, the primes of $\mathcal{O}$ containing p are exactly $\mathfrak{p}_i = (p, g_i(\tau))$, where $\bar{g} = \prod_i \bar{g}_i$ is the factorization into monic irreducibles over $\mathbb{F}_p$ and g_i lifts $\bar{g}_i$; moreover $\mathcal{O}/\mathfrak{p}_i \cong \mathbb{F}_p[x]/(\bar{g}_i)$, so $N(\mathfrak{p}_i) = p^{\deg \bar{g}_i}$. As $\bar{g}$ is a monic quadratic over a field, it is either a product of two distinct monic linear factors, irreducible, or the square of a linear factor; we claim these correspond to $\left(\frac{D}{p}\right) = 1, -1, 0$ respectively.

For p odd, complete the square: $\bar{g}(x) = \left(x - \frac{\bar{D}}{2}\right)^2 - \frac{\bar{D}}{4}$, so $\bar{g}$ has two distinct roots iff $\bar{D}$ is a nonzero square, a double root iff $\bar{D} = 0$, and is irreducible iff $\bar{D}$ is a nonsquare — matching the Legendre symbol. For $p = 2$, write $c := \frac{D(D-1)}{4}$ for the constant term of g ; the middle coefficient reduces to $\bar{D}$. If D is even, then $c \equiv \frac{D}{4} \pmod{2}$ ($D-1$ odd) and $\bar{g} = x^2 + \bar{c} = (x + \bar{c})^2$ — a double root. If $D \equiv 1 \pmod{8}$, then $\frac{D-1}{4}$ is even, so c is even and $\bar{g} = x^2 + x = x(x+1)$ — two distinct roots. If $D \equiv 5 \pmod{8}$, then $\frac{D-1}{4}$ and D are odd, so c is odd and $\bar{g} = x^2 + x + 1$, which has no root in $\mathbb{F}_2$ and is irreducible. Since an odd fundamental discriminant is $\equiv 1 \pmod{4}$, the residues 1, 5 mod 8 exhaust the odd case.

Step 2: splitting behavior and norms. If $\bar{h} \mid \bar{g}$ with complementary factor $\tilde{\bar{h}}$ and integral lifts $h, \tilde{h}$, then $g = h\tilde{h} + pr$ for some $r \in \mathbb{Z}[x]$, so evaluating at τ ,

$$h(\tau)\tilde{h}(\tau) = -pr(\tau) \in p\mathcal{O}. \quad (*)$$

Split case $\left(\frac{D}{p}\right) = 1$: $\bar{g} = (x - \bar{a})(x - \bar{b})$ with $\bar{a} \neq \bar{b}$, so the primes above p are $\mathfrak{p}_1 = (p, \tau - a)$ and $\mathfrak{p}_2 = (p, \tau - b)$, each of norm p . They are distinct: otherwise their common value contains both p and $(\tau - a) - (\tau - b) = b - a$ with $p \nmid b - a$, hence contains 1. Moreover $\mathfrak{p}_1\mathfrak{p}_2$ is generated by $p^2, p(\tau - a), p(\tau - b)$ and $(\tau - a)(\tau - b) \in p\mathcal{O}$ by $(*)$, so $\mathfrak{p}_1\mathfrak{p}_2 \subseteq p\mathcal{O}$; both sides have index p^2 ($N(\mathfrak{p}_1\mathfrak{p}_2) = N(\mathfrak{p}_1)N(\mathfrak{p}_2)$ by toolkit (iii), and $[\mathcal{O} : p\mathcal{O}] = p^2$ since $\mathcal{O} \cong \mathbb{Z}^2$ additively), so $p\mathcal{O} = \mathfrak{p}_1\mathfrak{p}_2$ and p splits.

Inert case $\left(\frac{D}{p}\right) = -1$: $\bar{g}$ irreducible, so the unique prime above p is $(p, g(\tau)) = p\mathcal{O}$ (as $g(\tau) = 0$), with $\mathcal{O}/p\mathcal{O} \cong \mathbb{F}_p[x]/(\bar{g}) \cong \mathbb{F}_{p^2}$ and $N(p\mathcal{O}) =$

p^2 ; p is inert.

Ramified case $\left(\frac{D}{p}\right) = 0$: $\bar{g} = (x - \bar{a})^2$, so the unique prime above p is $\mathfrak{p} = (p, \tau - a)$, of norm p ; by (*) with $h = \tilde{h} = x - a$ we get $\mathfrak{p}^2 = (p^2, p(\tau - a), (\tau - a)^2) \subseteq p\mathcal{O}$, and both sides have index p^2 (toolkit (iv)), so $p\mathcal{O} = \mathfrak{p}^2$ and p ramifies.

Step 3: reduction to primes above p . Let $N(\mathfrak{a}) = p^m$; then $\mathfrak{a} \neq 0$ by toolkit (ii), so $\mathfrak{a} = \mathfrak{q}_1^{a_1} \cdots \mathfrak{q}_t^{a_t}$ uniquely with $\mathfrak{q}_i$ distinct primes, $a_i \geq 1$. By toolkit (iii)–(iv), $p^m = \prod_i N(\mathfrak{q}_i)^{a_i}$, and by (v) each $N(\mathfrak{q}_i)$ is a power > 1 of the rational prime under $\mathfrak{q}_i$; unique factorization in $\mathbb{Z}$ forces every $\mathfrak{q}_i$ to lie above p . Conversely every product of primes above p has p -power norm, and distinct exponent vectors give distinct ideals. Hence the ideals of norm p^m biject with exponent vectors $(a_{\mathfrak{q}})_{\mathfrak{q}|p}$ satisfying $\prod_{\mathfrak{q}} N(\mathfrak{q})^{a_{\mathfrak{q}}} = p^m$. (For $m = 0$: the zero vector, i.e. $\mathfrak{a} = \mathcal{O}$, count 1.)

Step 4: counting. Split: two distinct primes of norm p ; solutions of $r + s = m$ with $r, s \geq 0$ give the $m + 1$ pairwise distinct ideals $\mathfrak{p}_1^r \mathfrak{p}_2^{m-r}$, $0 \leq r \leq m$. Inert: one prime of norm p^2 ; the equation $p^{2r} = p^m$ has one solution ($\mathfrak{a} = p^{m/2}\mathcal{O}$) if m is even, none if m is odd. Ramified: one prime $\mathfrak{p}$ of norm p ; the unique solution is $\mathfrak{a} = \mathfrak{p}^m$, for every m .

Step 5: uniform form. With $\chi := \left(\frac{D}{p}\right)$ and $0^0 = 1$: $\sum_{j=0}^m \chi^j$ equals $m + 1$ if $\chi = 1$; equals 1 (m even) or 0 (m odd) if $\chi = -1$; and equals 1 if $\chi = 0$ (only the $j = 0$ term). This matches Step 4 in every case. $\square$

Theorem 3.22 (Ideals of norm p^m in $\mathcal{O}_d$, $p \mid f$; corrected Theorem 3.1.2).

Lean: `QuadraticOrder.idealCount_prime_pow_of_le`, `QuadraticOrder.`

`idealCount_prime_pow_ramified`, `QuadraticOrder.`

`idealCount_prime_pow_inert`, `QuadraticOrder.idealCount_prime_pow_split`

Uses: §1.1, §2.1

Let $D < 0$ be a fundamental discriminant, $K = \mathbb{Q}(\sqrt{D})$ with maximal order $\mathcal{O}_K = \mathbb{Z}[(D + \sqrt{D})/2]$, let $f \geq 1$ and $d := Df^2$, and let $\mathcal{O}_d = \mathbb{Z}[\tau]$, $\tau := (d + \sqrt{d})/2$, be the order of discriminant d and conductor f , so that $\mathcal{O}_d \cong \mathbb{Z}[x]/(g)$ with $g(x) := x^2 - dx + \frac{d^2 - d}{4}$. Write $[\mathcal{O}_d : \mathfrak{a}]$ for the (finite) index of a nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_d$, i.e. its norm.

Let p be a prime with $p \mid f$, set $w := v_p(f) \geq 1$, let $m \geq 0$, and let $\varepsilon := 1$ if m is even and $\varepsilon := 0$ if m is odd. Define

$$v := v_p(d/4) := \begin{cases} v_p(d) & p \text{ odd}, \\ v_2(d) - 2 & p = 2 \end{cases} \quad (\text{legitimate: } 2 \mid f \Rightarrow 4 \mid d), \quad \text{so } v = \begin{cases} v_p(D) + 2w & p \text{ odd}, \\ v_2(D) + 2w - 2 & p = 2. \end{cases}$$

Then the number of ideals of norm p^m in $\mathcal{O}_d$ (invertible or not) is given as follows.

- If $v \geq m$:

$$\#\{\mathfrak{a} \subseteq \mathcal{O}_d : [\mathcal{O}_d : \mathfrak{a}] = p^m\} = \sum_{i=0}^{\lfloor m/2 \rfloor} p^i = \frac{p^{\lfloor m/2 \rfloor + 1} - 1}{p - 1}.$$

- If $v < m$:

$$\#\{\mathfrak{a} \subseteq \mathcal{O}_d : [\mathcal{O}_d : \mathfrak{a}] = p^m\} = \begin{cases} \sum_{i=0}^w p^i & p \text{ ramified in } \mathcal{O}_K \ (p \mid D), \\ \sum_{i=0}^{w+\varepsilon-1} p^i & p \text{ inert in } \mathcal{O}_K, \\ \sum_{i=0}^{w+\varepsilon-1} p^i + p^w (m + 1 - 2w - \varepsilon) & p \text{ split in } \mathcal{O}_K. \end{cases}$$

The statement is uniform in p : there is no separate display for $p = 2$.

Proof.

Uses: §3.18, §3.8

Write $r(p^m) := \#\{\mathfrak{a} \subseteq \mathcal{O}_d : [\mathcal{O}_d : \mathfrak{a}] = p^m\}$ and $\mu := m \bmod 2 = 1 - \varepsilon$.

Step 1 (reduction to root counting). By the normal-form bijection of Lemma 3.18 (in its corrected form, which applies to *all* ideals of index p^m — since $p \mid f$, the polynomial $g \bmod p$ is the square of a linear polynomial, so $\mathcal{O}_d/p\mathcal{O}_d$ is local, there is a unique prime of $\mathcal{O}_d$ above p , and every ideal of p -power index is primary for it — and which includes the minimal-integer normalization $\mathfrak{a}_{k,A} \cap \mathbb{Z} = p^k \mathbb{Z}$ preventing double counting), the ideals of index p^m are exactly the

$$\mathfrak{a}_{k,A} = p^k \mathbb{Z} \oplus p^{m-k}(\tau - A) \mathbb{Z}, \quad \lceil m/2 \rceil \leq k \leq m, \quad [A] \in \mathbb{Z}/p^{2k-m}, \quad g(A) \equiv 0 \bmod p^{2k-m},$$

each exactly once. Hence

$$r(p^m) = \sum_{k=\lceil m/2 \rceil}^m N_k, \quad N_k := \#\{A \in \mathbb{Z}/p^{2k-m} : g(A) \equiv 0 \bmod p^{2k-m}\}.$$

Set $s := 2k - m$; as k runs over its range, s runs over the integers of the same parity as m with $\mu \leq s \leq m$, and $k - \lceil m/2 \rceil = (s - \mu)/2 = \lfloor s/2 \rfloor$.

Step 2 (completing the square). In $\mathbb{Z}[\frac{1}{2}][x]$ one has $g(x) = (x - d/2)^2 - d/4$. If p is odd, 2 is invertible mod p^s and the substitution $x \mapsto x + d\bar{2}$ (with

$\bar{2} := 2^{-1} \bmod p^s$) is a bijection of $\mathbb{Z}/p^s$; if $p = 2$ then $4 \mid d$ and the identity is exact in $\mathbb{Z}[x]$. In both cases

$$N_k = \#\{y \in \mathbb{Z}/p^s : y^2 \equiv q \bmod p^s\}, \quad q := \begin{cases} d\bar{4} & p \text{ odd}, \\ d/4 & p = 2, \end{cases}$$

and any integer lift of q has p -adic valuation exactly v when $v < s$, while $q \equiv 0 \bmod p^s$ when $v \geq s$.

Step 3 (valuation dichotomy; the case $v \geq m$). Put $\kappa := (v + m)/2$, so $k \leq \kappa \iff s \leq v$. If $s \leq v$ then $y^2 \equiv 0 \bmod p^s$, whose solutions are exactly the multiples of $p^{\lceil s/2 \rceil}$, giving $N_k = p^{\lfloor s/2 \rfloor} = p^{k - \lceil m/2 \rceil}$. If $v \geq m$ every k in range satisfies $s \leq m \leq v$, so

$$r(p^m) = \sum_{k=\lceil m/2 \rceil}^m p^{k - \lceil m/2 \rceil} = \sum_{i=0}^{\lfloor m/2 \rfloor} p^i,$$

proving the first branch (this also covers $m = 0$). Assume $v < m$ from now on.

Step 4 (the sub- κ geometric block). The terms with $k \leq \kappa$ total $\Sigma_0 := \sum_{i=0}^{i_0} p^i$ with $i_0 := \lfloor (v + m)/2 \rfloor - \lceil m/2 \rceil$ (an empty sum, $= 0$, if $i_0 < 0$). A direct computation gives: $i_0 = (v - 1)/2$ if v is odd (both parities of m), and $i_0 = v/2 - 1 + \varepsilon$ if v is even.

Step 5 (beyond κ). The remaining terms have $s \in S := \{s : v < s \leq m, s \equiv m \bmod 2\}$, which is nonempty; its least element s_1 is $v + 1$ or $v + 2$.

(a) v odd — exactly the cases p odd with $p \mid D$ (then $v_p(D) = 1$, $v = 2w + 1$) and $p = 2$ with $D \equiv 8 \bmod 16$ ($v = 2w + 1$); both ramified. Then $N_k = 0$ for all $s > v$: a solution would force $v_p(y^2) = v$ by the ultrametric inequality (any integer lift of q has valuation $v < s$), impossible since $v_p(y^2)$ is even. Hence $r(p^m) = \Sigma_0 = \sum_{i=0}^w p^i$, the ramified formula.

(b) v even. Write $r := v/2$ and choose an integer u coprime to p with $q \equiv p^{2r}u \bmod p^s$. Concretely: for p odd unramified, $r = w$ and $u \equiv D(f/p^w)^2 \bar{4} \bmod p$, so the Legendre symbol satisfies $(\frac{u}{p}) = (\frac{D}{p})$; for $p = 2$ with D odd, $r = w - 1$ and, writing $f = 2^w f'$ with f' odd, $u = Df'^2 \equiv D \bmod 8$; for $p = 2$ with $D \equiv 4 \bmod 8$, $r = w$ and $u = (D/4)f'^2 \equiv 3 \bmod 4$. Since $2r = v < s$ on S , Lemma 3.8 (corrected form) applies: for p odd there are $2p^r$ solutions if $(\frac{u}{p}) = 1$ and none otherwise; for $p = 2$ there are 2^r solutions if $s - 2r = 1$ (unconditionally), 2^{r+1} if $s - 2r = 2$ and $u \equiv 1 \bmod 4$, 2^{r+2} if $s - 2r \geq 3$ and $u \equiv 1 \bmod 8$, and none otherwise. **The case distinction is governed by $s - 2r$, not by s ;** failing to track this is precisely the error in the thesis proof (ERRATA E3).

(b1) *p odd inert* ($(\frac{D}{p}) = -1$): no contributions, and $r(p^m) = \Sigma_0 = \sum_{i=0}^{w-1+\varepsilon} p^i$, the inert formula ($v = 2w$).

(b2) *p odd split* ($(\frac{D}{p}) = 1$): every $s \in S$ contributes $2p^w$. Counting, $\#S = (m+1-2w-\varepsilon)/2 \geq 1$, so

$$r(p^m) = \sum_{i=0}^{w-1+\varepsilon} p^i + 2p^w \cdot \frac{m+1-2w-\varepsilon}{2} = \sum_{i=0}^{w+\varepsilon-1} p^i + p^w(m+1-2w-\varepsilon),$$

the split formula.

(b3) $p = 2$, $D \equiv 4 \pmod{8}$ (ramified, $v = 2w$, $r = w$, $u \equiv 3 \pmod{4}$): the only admissible value is $s - 2r = 1$, i.e. $s = 2w + 1 \in S$ iff m is odd (and then $2w + 1 \leq m$ since $v = 2w < m$), contributing 2^w ; the cases $s - 2r = 2$ and $s - 2r \geq 3$ are excluded by $u \equiv 3 \pmod{4}$. Hence

$$r(2^m) = \sum_{i=0}^{w-1+\varepsilon} 2^i + (1-\varepsilon)2^w = \sum_{i=0}^w 2^i$$

for both parities of m (for m even this is Σ_0 alone; for m odd it is $(2^w - 1) + 2^w$), the ramified formula. This corrects ERRATA E2: the thesis claimed $2^{w+\varepsilon} - 1$, wrong for m odd.

(b4) $p = 2$, D odd ($v = 2w - 2$, $r = w - 1$, $u \equiv D \pmod{8}$; here $s - 2r = s - 2w + 2$). By Lemma 3.8: $s = 2w - 1$ ($\in S$ iff m odd) contributes 2^{w-1} unconditionally; $s = 2w$ ($\in S$ iff m even) contributes 2^w since $u \equiv D \equiv 1 \pmod{4}$; and each $s \geq 2w + 1$ in S contributes 2^{w+1} if $D \equiv 1 \pmod{8}$ (2 split) and 0 if $D \equiv 5 \pmod{8}$ (2 inert). In particular the least element s_1 of S contributes exactly $2^{w-1+\varepsilon}$ for every $w \geq 1$ — the term the thesis found only at $w = 1$ (ERRATA E1/E3). Therefore, with $\Sigma_0 = \sum_{i=0}^{w-2+\varepsilon} 2^i = 2^{w-1+\varepsilon} - 1$ (empty when $w = 1$, m odd):

$$\text{inert: } r(2^m) = (2^{w-1+\varepsilon} - 1) + 2^{w-1+\varepsilon} = \sum_{i=0}^{w+\varepsilon-1} 2^i;$$

$$\text{split: } r(2^m) = (2^{w-1+\varepsilon} - 1) + 2^{w-1+\varepsilon} + 2^{w+1} \cdot \frac{m+1-2w-\varepsilon}{2} = \sum_{i=0}^{w+\varepsilon-1} 2^i + 2^w(m+1-2w-\varepsilon),$$

where the number of $s \in S$ with $s \geq 2w + 1$ is $(m+1-2w-\varepsilon)/2 \geq 0$, exactly as in (b2).

Cases (a) and (b3) together are exactly “ p ramified”; (b1) and the inert half of (b4) are “ p inert”; (b2) and the split half of (b4) are “ p split”. Every total matches the claimed p -uniform display. $\square$

Divergence from the thesis. This node states the corrected form of Theorem 3.1.2 (ERRATA E1, E2, E4): the thesis's 2-adic inert branch for $w \neq 1$ ($2^{w-1+\varepsilon} - 1$) is replaced by $2^{w+\varepsilon} - 1$ for all $w \geq 1$ (E1); the 2-adic ramified branch with $D \equiv 4 \pmod{8}$ ($2^{w+\varepsilon} - 1$) is replaced by $2^{w+1} - 1$ for both parities of m (E2); after these corrections the five-case 2-adic display coincides with the odd- p formulas at $p = 2$ and is therefore dissolved into the single p -uniform statement above (E4). Sanity anchors (verified by explicit enumeration; ERRATA E1/E2): $d = -16, m = 3 \mapsto 3$; $d = -48, m = 3 \mapsto 3$ and $m = 4 \mapsto 7$; $d = -192, m = 5 \mapsto 7$. These are enforced at build time as `native_decide` regression vectors in `QuadraticOrder/Harness/Regression.lean`.

Corollary 3.23 (Corrected Corollary 3.1.3).

Lean: `QuadraticOrder.invertibleIdealCount_prime_pow_trivial`,
`QuadraticOrder.invertibleIdealCount_prime_pow_ramified`,
`QuadraticOrder.invertibleIdealCount_prime_pow_inert`, `QuadraticOrder.invertibleIdealCount_prime_pow_split`

Uses: §1.1, §3.18, §3.19, §3.20, §2.1, §2.13

Let $d = Df^2$, where $D < 0$ is fundamental, and let the prime p divide f . Put $w = v_p(f)$ and put $v = v_p(d)$ for odd p , while $v = v_2(d) - 2 = v_2(d/4)$ for $p = 2$. The number $C_d^\times(p^m)$ of invertible ideals of $\mathcal{O}_d$ of index p^m is as follows.

If $m < v$, or if $m = v = 2w - 2$, then

$$C_d^\times(p^m) = \begin{cases} p^{m/2}, & m \text{ even}, \\ 0, & m \text{ odd}. \end{cases}$$

If $v < m$, or if $m = v$ and $2w \leq v$, then

$$C_d^\times(p^m) = \begin{cases} p^w, & (D/p) = 0, \\ p^w + p^{w-1}, & (D/p) = -1 \text{ and } m \text{ even}, \\ 0, & (D/p) = -1 \text{ and } m \text{ odd}, \\ (p^w - p^{w-1})(m + 1 - 2w), & (D/p) = 1. \end{cases}$$

In particular the boundary $m = v = 2w - 2$, which occurs for $p = 2$ and odd D , belongs to the first regime (ERRATA E5).

Proof.

Uses: §3.18, §3.19, §3.20, §2.1, §2.13

Write $g(X) = X^2 - dX + (d^2 - d)/4$. The normal-form bijection writes every ideal of index p^m uniquely as

$$\mathfrak{a}_{k,A} = p^k \mathbb{Z} \oplus p^{m-k}(\tau - A)\mathbb{Z}, \text{quad}[m/2] \leq k \leq m,$$

where, for $s = 2k - m$, the parameter A is a root of g modulo p^s . Put $r = m - k$. Lemma 3.19 says that the contraction of the local principal ideal generated by $p^r(\tau - A)$ has parameters $r + \nu, 2r + \nu$, with $\nu = v_p(g(A))$. Uniqueness and Lemma 3.20 therefore give, for $s > 0$,

$$\mathfrak{a}_{k,A} \text{ invertible} \iff v_p(g(A)) = s. \quad (1)$$

For $s = 0$ the ideal is the principal ideal $p^{m/2}\mathcal{O}_d$.

Let E_s count the exact roots in (1), and set $E_0 = 1$. Thus

$$C_d^\times(p^m) = \sum_{\substack{0 \leq s \leq m \\ s \equiv m \pmod{2}}} E_s. \quad (2)$$

Completing the square translates g to $x^2 - q$, where $v_p(q) = v$ (for $p = 2$, $q = d/4$ is integral). If $0 < s < v$, exact valuation forces $s = 2t$ and $x = p^t y$ with y a unit; hence

$$E_{2t} = p^t - p^{t-1}, \quad E_{2t+1} = 0. \quad (3)$$

The even terms telescope to $1 + \sum_{t=1}^h (p^t - p^{t-1}) = p^h$, proving the first regime.

Write $q = p^v u$. At and above the boundary, reducing $y^2 - u$ after extracting $p^{\lfloor v/2 \rfloor}$ gives the following exhaustive alternatives. If $v = 2w + 1$, the boundary contributes p^w and an odd-valuation unit cannot be a square above it. If $v = 2w$ and p is odd, the sum through the boundary is $p^w + p^{w-1}$ for nonsquare u and $p^w - p^{w-1}$ for square u ; the former has no tail, while every permitted tail exponent in the latter case contributes $2(p^w - p^{w-1})$. If $p = 2$ and $v = 2w$ with $u \equiv 3 \pmod{4}$, the even boundary and the possible odd boundary each give 2^w , and there is no later root. Finally, if $p = 2$, D is odd, and $v = 2w - 2$, the even sum through v is 2^{w-1} ; for $u \equiv 5 \pmod{8}$ only exponent $v + 2$ contributes (with 2^w classes), while for $u \equiv 1 \pmod{8}$ the first two tail exponents contribute zero and every permitted exponent from $v + 3$ onward contributes 2^w .

These counts use only that odd squares are 1 modulo 8 and that, for odd p , a square unit has two simple roots while a nonsquare has none. Inserting them into (2) gives p^w in the ramified case, $p^w + p^{w-1}$ for even inert m and zero for odd inert m , and

$$(p^w - p^{w-1})(m + 1 - 2w)$$

in the split case. The $v = 2w - 2$ equality remains in the telescoping block, which is exactly the corrected boundary asserted above. $\square$

Remark 3.24.

Uses: §3.23

The thesis places every equality $m = v$ in the second regime. At $d = -112$, $w = 2$, $m = v = 2$, this makes its split expression negative; the correct count is 2. The regression rows for $d = -48, -112, -192$ enforce the corrected boundary.

Proposition 3.25 (Prime-power factorization of ideal counts).

Lean: `QuadraticOrder.idealCount_eq_prod_primeFactors`, `QuadraticOrder.invertibleIdealCount_eq_prod_primeFactors`

Uses: §1.1, §2.1

Let d be an imaginary quadratic discriminant, $\mathcal{O} = \mathcal{O}_d$, and $n \geq 1$. Sending an ideal to its family of primary components is a bijection

$$\{\mathfrak{a} \subseteq \mathcal{O} : [\mathcal{O} : \mathfrak{a}] = n\} \longleftrightarrow \prod_{p|n} \{\mathfrak{b} \subseteq \mathcal{O} : [\mathcal{O} : \mathfrak{b}] = p^{v_p(n)}\},$$

under which $\mathfrak{a}$ is invertible if and only if every component is invertible. Consequently

$$r_d(n) = \prod_{p|n} r_d(p^{v_p(n)}), \quad r_d^\times(n) = \prod_{p|n} r_d^\times(p^{v_p(n)}).$$

Proof.

Uses: §1.1, §2.1

We use the following index facts from the index layer, each elementary: (0.1) an ideal of finite index N contains N (Lagrange in $\mathcal{O}/\mathfrak{a}$); (0.2) if $\mathfrak{b} \subseteq \mathfrak{c}$ then $[\mathcal{O} : \mathfrak{c}] \mid [\mathcal{O} : \mathfrak{b}]$; (0.3) ideals of coprime finite indices are comaximal (by (0.2), the index of the sum divides both indices); (0.4) $[\mathcal{O} : \mathfrak{b}\mathfrak{c}]$ is finite and divides $N_{\mathfrak{b}}^2 N_{\mathfrak{c}}$, since $N_{\mathfrak{b}}\mathfrak{c} \subseteq \mathfrak{b}\mathfrak{c}$ by (0.1) and $[\mathcal{O} : N_{\mathfrak{b}}\mathfrak{c}] = N_{\mathfrak{b}}^2 N_{\mathfrak{c}}$; (0.5) comaximal ideals satisfy $\mathfrak{b} \cap \mathfrak{c} = \mathfrak{b}\mathfrak{c}$ (write $1 = b + c$ and expand $x = xb + xc$); (0.6) the Chinese remainder theorem for finitely many pairwise comaximal ideals.

Components. Let $[\mathcal{O} : \mathfrak{a}] = n$ and $M := \mathcal{O}/\mathfrak{a}$, a finite $\mathcal{O}$ -module of cardinality n . As a finite abelian group, $M = \bigoplus_{p|n} M_p$ with M_p the p -primary component, of order $p^{v_p(n)}$. Each M_p is an $\mathcal{O}$ -submodule: multiplication by $y \in \mathcal{O}$ is an additive endomorphism, and $p^k x = 0$ implies $p^k(yx) = y(p^k x) = 0$. Hence the projections $\pi_p : M \rightarrow M_p$ are $\mathcal{O}$ -linear, and

$$\mathfrak{a}_p := \ker(\mathcal{O} \twoheadrightarrow M \xrightarrow{\pi_p} M_p)$$

is an ideal containing $\mathfrak{a}$ with $\mathcal{O}/\mathfrak{a}_p \cong M_p$, so $[\mathcal{O} : \mathfrak{a}_p] = p^{v_p(n)}$; and $\bigcap_{p|n} \mathfrak{a}_p = \ker(\mathcal{O} \rightarrow \bigoplus_p M_p = M) = \mathfrak{a}$.

Inverse map. Given ideals $\mathfrak{b}_p$ with $[\mathcal{O} : \mathfrak{b}_p] = p^{v_p(n)}$, they are pairwise comaximal by (0.3), so by (0.6) the intersection $\mathfrak{b} := \bigcap_p \mathfrak{b}_p$ satisfies $\mathcal{O}/\mathfrak{b} \cong \prod_p \mathcal{O}/\mathfrak{b}_p$ and has index n . The p -primary component of the product is the factor $\mathcal{O}/\mathfrak{b}_p$ (a p -group; the other factors have order prime to p), so the component construction applied to $\mathfrak{b}$ returns exactly the $\mathfrak{b}_p$. The two constructions are therefore mutually inverse, and counting gives the product formula for r_d .

Invertibility. By (0.5) and induction, $\mathfrak{a} = \bigcap_p \mathfrak{a}_p = \prod_p \mathfrak{a}_p$. If each $\mathfrak{a}_p$ has a fractional inverse $\mathfrak{c}_p$, then $\mathfrak{a} \cdot \prod_p \mathfrak{c}_p = \mathcal{O}$. Conversely, if $\mathfrak{a}\mathfrak{c} = \mathcal{O}$, then for each p , $\mathfrak{a}_p \cdot (\mathfrak{c} \prod_{q \neq p} \mathfrak{a}_q) = \mathcal{O}$. So the bijection restricts to invertible ideals on both sides, giving the product formula for $r_d^\times$. $\square$

Lemma 3.26 (Conductor-avoiding extension and contraction).

Uses: §1.1, §2.1

Let $d = Df^2$ and let $N \geq 1$ with $\gcd(N, f) = 1$. Then:

1. $\mathcal{O}_d \cap N\mathcal{O}_K = N\mathcal{O}_d$, and the inclusion $\mathcal{O}_d \hookrightarrow \mathcal{O}_K$ induces a ring isomorphism $\mathcal{O}_d/N\mathcal{O}_d \xrightarrow{\sim} \mathcal{O}_K/N\mathcal{O}_K$;
2. for every ideal $\mathfrak{c} \subseteq \mathcal{O}_d$ of index N : $\mathfrak{c}\mathcal{O}_K = \mathfrak{c} + N\mathcal{O}_K$, $\mathfrak{c}\mathcal{O}_K \cap \mathcal{O}_d = \mathfrak{c}$, and $[\mathcal{O}_K : \mathfrak{c}\mathcal{O}_K] = N$.

Proof.

Uses: §1.1, §2.1

Fix $\alpha, \beta \in \mathbb{Z}$ with $1 = \alpha N + \beta f$, and recall $f\mathcal{O}_K \subseteq \mathcal{O}_d = \mathbb{Z} + f\mathcal{O}_K$.

(1) If $x = Ny \in \mathcal{O}_d$ with $y \in \mathcal{O}_K$, then $y = \alpha Ny + \beta fy = \alpha x + \beta(fy) \in \mathcal{O}_d$ since $fy \in f\mathcal{O}_K \subseteq \mathcal{O}_d$; hence $x \in N\mathcal{O}_d$, proving $\mathcal{O}_d \cap N\mathcal{O}_K \subseteq N\mathcal{O}_d$ (the reverse inclusion is clear). The composite $\mathcal{O}_d \hookrightarrow \mathcal{O}_K \rightarrow \mathcal{O}_K/N\mathcal{O}_K$ is a ring homomorphism with kernel $\mathcal{O}_d \cap N\mathcal{O}_K = N\mathcal{O}_d$, so the induced map is injective; it is surjective because every $z \in \mathcal{O}_K$ satisfies $z = \alpha Nz + \beta fz \equiv \beta fz \pmod{N\mathcal{O}_K}$ with $\beta fz \in \mathcal{O}_d$.

(2) First, $N \in \mathfrak{c}$ (Lagrange), so $N\mathcal{O}_K \subseteq \mathfrak{c}\mathcal{O}_K$ and $\mathfrak{c} + N\mathcal{O}_K \subseteq \mathfrak{c}\mathcal{O}_K$; conversely each generator cz ($c \in \mathfrak{c}$, $z \in \mathcal{O}_K$) satisfies $cz = \alpha N(cz) + \beta c(fz) \in N\mathcal{O}_K + \mathfrak{c}$ because $fz \in \mathcal{O}_d$ makes $c(fz) \in \mathfrak{c}$. This proves $\mathfrak{c}\mathcal{O}_K = \mathfrak{c} + N\mathcal{O}_K$. Next, if $x \in \mathfrak{c}\mathcal{O}_K \cap \mathcal{O}_d$, write $x = c + Ny$ with $c \in \mathfrak{c}$, $y \in \mathcal{O}_K$; then $Ny = x - c \in \mathcal{O}_d \cap N\mathcal{O}_K = N\mathcal{O}_d \subseteq \mathfrak{c}$ by (1) and Lagrange, so $x \in \mathfrak{c}$, proving $\mathfrak{c}\mathcal{O}_K \cap \mathcal{O}_d = \mathfrak{c}$. Finally, under the isomorphism of (1) the image of $\mathfrak{c}/N\mathcal{O}_d$ is $(\mathfrak{c} + N\mathcal{O}_K)/N\mathcal{O}_K = \mathfrak{c}\mathcal{O}_K/N\mathcal{O}_K$, so $[\mathcal{O}_K : \mathfrak{c}\mathcal{O}_K] = [\mathcal{O}_d/N\mathcal{O}_d : \mathfrak{c}/N\mathcal{O}_d] = N$. $\square$

Theorem 3.27 (Conductor-avoiding correspondence).

Lean: `QuadraticOrder.idealCount_eq_of_coprime_conductor`

Uses: §1.1, §2.1

Let $d = Df^2$ and let $n \geq 1$ with $\gcd(n, f) = 1$. Then extension $\mathfrak{a} \mapsto \mathfrak{a}\mathcal{O}_K$ and contraction $\mathfrak{A} \mapsto \mathfrak{A} \cap \mathcal{O}_d$ are mutually inverse, index-preserving bijections

$$\{\mathfrak{a} \subseteq \mathcal{O}_d : [\mathcal{O}_d : \mathfrak{a}] = n\} \longleftrightarrow \{\mathfrak{A} \subseteq \mathcal{O}_K : [\mathcal{O}_K : \mathfrak{A}] = n\},$$

and every ideal of $\mathcal{O}_d$ of index coprime to f is invertible. In particular

$$r_d(n) = r_d^\times(n) = r_D(n) = r_D^\times(n).$$

Proof.

Uses: §3.26, §1.1, §2.1

Bijection. Every ideal of index n contains n (Lagrange), so ideals of index n in $\mathcal{O}_d$ (resp. $\mathcal{O}_K$) correspond to ideals of $\mathcal{O}_d/n\mathcal{O}_d$ (resp. $\mathcal{O}_K/n\mathcal{O}_K$) with quotient of cardinality n . The ring isomorphism of Lemma 3.26(1) (with $N = n$) matches the two collections bijectively, preserving indices. By Lemma 3.26(2) the forward map is $\mathfrak{a} \mapsto \mathfrak{a}\mathcal{O}_K$ (of index n), and the inverse is $\mathfrak{A} \mapsto \mathfrak{A} \cap \mathcal{O}_d$ (the preimage of $\mathfrak{A}/n\mathcal{O}_K$ in $\mathcal{O}_d$, using $n\mathcal{O}_K \subseteq \mathfrak{A}$); mutual inversion is $\mathfrak{c}\mathcal{O}_K \cap \mathcal{O}_d = \mathfrak{c}$ together with bijectivity of the mod- n isomorphism.

Invertibility. Let $[\mathcal{O}_d : \mathfrak{a}] = n$ and $\mathfrak{A} := \mathfrak{a}\mathcal{O}_K$. In the maximal order, $\mathfrak{A}\bar{\mathfrak{A}} = N(\mathfrak{A})\mathcal{O}_K = n\mathcal{O}_K$ (standard; by multiplicativity this reduces to prime ideals, where it is checked in each splitting case — cf. Cox, Lemma 7.14). Conjugation is a ring automorphism of $\mathcal{O}_d$ ($\bar{\tau}_d = d - \tau_d$), so $\bar{\mathfrak{a}}$ is an ideal of index n , and $[\mathcal{O}_d : \mathfrak{a}\bar{\mathfrak{a}}]$ is finite and divides n^3 , hence is coprime to f . Applying Lemma 3.26(2) to $\mathfrak{c} = \mathfrak{a}\bar{\mathfrak{a}}$, and using that extension is multiplicative and commutes with conjugation,

$$\mathfrak{a}\bar{\mathfrak{a}} = (\mathfrak{a}\bar{\mathfrak{a}})\mathcal{O}_K \cap \mathcal{O}_d = \mathfrak{A}\bar{\mathfrak{A}} \cap \mathcal{O}_d = n\mathcal{O}_K \cap \mathcal{O}_d = n\mathcal{O}_d,$$

the last step by Lemma 3.26(1). Hence $\mathfrak{a} \cdot \frac{1}{n}\bar{\mathfrak{a}} = \mathcal{O}_d$ and $\mathfrak{a}$ is invertible.

Counts. The bijection gives $r_d(n) = r_D(n)$; invertibility of every ideal in sight gives $r_d^\times(n) = r_d(n)$, and $\mathcal{O}_K$ is Dedekind, so $r_D^\times(n) = r_D(n)$. $\square$

Theorem 3.28 (Ideal counts for arbitrary norm: assembled closed form).

Uses: §1.1

Let $d = Df^2$ be an imaginary quadratic discriminant. For a prime p and $m \geq 0$ define local factors as follows. If $p \nmid f$:

$$t_p(m) = u_p(m) = \begin{cases} m+1 & p \text{ split in } \mathcal{O}_K, \\ \varepsilon(m) & p \text{ inert in } \mathcal{O}_K, \\ 1 & p \mid D. \end{cases}$$

If $p \mid f$, with $w = v_p(f)$ and $v = v_p(d/4)$:

$$t_p(m) = \begin{cases} \frac{p^{\lfloor m/2 \rfloor + 1} - 1}{p - 1} & m \leq v, \\ \frac{p^{w+1} - 1}{p - 1} & m > v, \ p \mid D, \\ \frac{p^{w+\varepsilon(m)} - 1}{p - 1} & m > v, \ p \text{ inert}, \\ \frac{p^{w+\varepsilon(m)} - 1}{p - 1} + p^w(m + 1 - 2w - \varepsilon(m)) & m > v, \ p \text{ split}, \end{cases}$$

$$u_p(m) = \begin{cases} \varepsilon(m) p^{m/2} & m < v, \text{ or } m = v = 2w - 2, \\ p^w & (m > v \text{ or } m = v \geq 2w), \ p \mid D, \\ \varepsilon(m)(p^w + p^{w-1}) & (m > v \text{ or } m = v \geq 2w), \ p \text{ inert}, \\ (p^w - p^{w-1})(m + 1 - 2w) & (m > v \text{ or } m = v \geq 2w), \ p \text{ split}. \end{cases}$$

(Read $\varepsilon(m) p^{m/2}$ as $p^{m/2}$ for m even and 0 for m odd. For $p \mid f$ one has $v \in \{2w, 2w + 1\}$ for p odd and $v \in \{2w - 2, 2w, 2w + 1\}$ for $p = 2$, with $v = 2w - 2$ iff D odd; so the two regimes of u_p are exhaustive and mutually exclusive.) Then for every $n \geq 1$,

$$r_d(n) = \prod_{p \mid n} t_p(v_p(n)), \quad r_d^\times(n) = \prod_{p \mid n} u_p(v_p(n));$$

equivalently the products may run over all primes, since $t_p(0) = u_p(0) = 1$.

Proof.

Uses: §3.25, §3.27, §3.21, §3.22, §3.23

Write $m_p := v_p(n)$. Proposition 3.25 factors both counts over the primes dividing n : $r_d(n) = \prod_{p \mid n} r_d(p^{m_p})$ and $r_d^\times(n) = \prod_{p \mid n} r_d^\times(p^{m_p})$.

For $p \nmid f$, $\gcd(p^{m_p}, f) = 1$, so Theorem 3.27 gives $r_d(p^{m_p}) = r_d^\times(p^{m_p}) = r_D(p^{m_p})$, and Proposition 3.21 evaluates $r_D(p^m)$ as $m + 1$ (p split), $\varepsilon(m)$ (p inert), 1 (p ramified); this is $t_p(m_p) = u_p(m_p)$.

For $p \mid f$, the corrected Theorem 3.22 (statement per ERRATA E4, p -uniform) evaluates $r_d(p^{m_p}) = t_p(m_p)$, and the corrected Corollary 3.23 (regime per ERRATA E5) evaluates $r_d^\times(p^{m_p}) = u_p(m_p)$. The parenthetical trichotomy of v follows from $d = Df^2$ with D fundamental: for odd $p \mid f$, $v = 2w + v_p(D)$ with $v_p(D) \in \{0, 1\}$; for $p = 2 \mid f$, $v = 2w - 2 + v_2(D)$ with $v_2(D) \in \{0, 2, 3\}$; in particular $v = 2w - 1$ never occurs, so at $m = v$ exactly one of $v = 2w - 2$, $v \geq 2w$ holds.

Finally, $t_p(0) = u_p(0) = 1$ for every prime p : if $p \nmid f$ all three cases give 1 at $m = 0$; if $p \mid f$ then $0 = m \leq v$ gives $t_p(0) = (p - 1)/(p - 1) = 1$, and for u_p either $v > 0$ (trivial branch, value $\varepsilon(0)p^0 = 1$) or $v = 0$, which forces $p = 2$, D odd, $w = 1$, i.e. $m = v = 2w - 2$ (again the trivial branch, value 1). Substituting the factor evaluations completes the proof. $\square$

Remark 3.29.

Uses: §3.28

(i) *Divergence from the thesis.* None of the above is stated in the thesis, which counts only prime-power norms; and the local factors at $p \mid f$ are the ERRATA-corrected statements (E1–E4 for Theorem 3.22, E5 for Corollary 3.23), not the thesis’s originals, two of whose 2-adic branches are provably false. The thesis’s Corollary 3.1.3 display also reads like a total count but means *invertible* ideals (ERRATA E6.1); the notation r_d vs. $r_d^\times$ removes the ambiguity. (ii) *Index is not multiplicative in $\mathcal{O}_d$.* For $d = -12$, $\mathfrak{p} = (2, 1 + \sqrt{-3})$ has index 2 but $\mathfrak{p}^2$ has index 8; hence Proposition 3.25 is proved via comaximal CRT decomposition, never via norm multiplicativity of products (and Mathlib’s Dedekind-gated `Ideal.absNorm` API is unusable here). (iii) *Coprimality is necessary* in Theorem 3.27: $r_{-12}(2) = 1$ while $r_{-3}(2) = 0$. (iv) *Lean route.* The mod- N comparison of Lemma 3.26 is where Mathlib’s conductor machinery applies (`conductor`, `Localization.localRingHom_bijective_of_not_conductor_le`, `comap_map_eq_map_adjoin_of_co`); a per-prime localization proof is a viable alternative to the elementary mod- N argument given here. (v) For $\gcd(n, f) = 1$ the closed form collapses to the classical divisor sum $r_d(n) = \sum_{c \mid n} \chi_D(c)$. (vi) *Role.* This theorem is the counting instrument for the Question 3.0.1 falsification loop: candidate reformulations of the Lauter–Viray count $A(N)$ for $\gcd(m, f_1) > 1$ are tested numerically against $v_\ell(F(m))$, which requires exact invertible-ideal counts of arbitrary composite norm, including norms meeting the conductor.

Chapter 4

Singular moduli: the statement layer

Placeholder for WP-I: the transcribed Gross–Zagier, Dorman, and Lauter–Viray statements, the CM axiom layer, and Conjecture 1.7 of Lauter–Viray. Fragments land in **results/** as the transcription pass (from **papers/**) completes; see **WORKPLAN.md** WP-I.